

A temporally restricted function of the Dopamine receptor Dop1R2 during memory formation

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The authors design and implement an elegant strategy to delete genomic sequences encoding the dopamine receptor dop1R2 from specific subsets of mushroom body neurons (ab, a'b' and gamma) and show that while none of these manipulations affect short term appetitive or aversive memory, loss of dop1R2 from ab or a'b' block the ability of flies to display measurable forms of longer forms of memory. These findings are **important** in confirming and extending prior observations, and well supported by **convincing** evidence that build on precise techniques for genetic perturbation.

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Abstract

Summary

Dopamine is a crucial neuromodulator involved in many brain processes, including learning and the formation of memories. Dopamine acts through multiple receptors and controls an intricate signaling network to regulate different tasks. While the diverse functions of dopamine are intensely studied, the interplay and role of the distinct dopamine receptors to regulate different processes is less well understood. An interesting candidate is the dopamine receptor Dop1R2 (also known as Damb), as it could connect to different downstream pathways. Dop1R2 is reported to be involved in forgetting and memory maintenance, however, the circuits requiring the receptors are unknown. To study Dop1R2 and its role in specific spatial and temporal contexts, we generated a conditional knock-out line using the CRISPR-Cas9 technique. Two FRT sites were inserted, allowing flippase-mediated excision of the dopamine receptor in neurons of interest. To study the function of Dop1R2, we knocked it out conditionally in the Mushroom body of *Drosophila melanogaster*, a well-studied brain region for memory formation. We show that Dop1R2 is required for later memory forms but not for short-term aversive or appetitive memories. Moreover, Dop1R2 is specifically required in the α/β -lobe and the α'/β' -lobe but not in the γ -lobe of the Mushroom body. Our findings show a spatially and temporally restricted role of Dop1R2 in the process of memory

formation highlighting the differential requirement of receptors during distinct phases of learning.

Introduction

The neuromodulator dopamine is involved in a plethora of brain functions. Among them are learning and memory, reinforcement signaling, reward, arousal, and motor functions (Klein et al., 2019; Missale et al., 1998; Tritsch & Sabatini, 2012). Perturbations in the dopaminergic system are associated with diseases like Parkinson's disease, addiction, depression, schizophrenia, and many more (Klein et al., 2019; Missale et al., 1998; Tritsch & Sabatini, 2012). Therefore, it is important to understand dopamine signaling in greater detail allowing to develop more effective therapeutics or preventive measures.

Dopamine is synthesized from tyrosine in dopaminergic neurons and binds to G-protein coupled receptors (GPCRs) (Klein et al., 2019; Missale et al., 1998; Yamamoto & Seto, 2014). These dopamine receptors have seven transmembrane domains, with an intracellular C-terminus and an extracellular N-terminus. The receptors interact with different G-proteins, which are heterotrimeric protein complexes consisting of an α -, β -, and γ -subunit. The G-protein complex binds to the third intracellular loop and the C-terminus of the receptors. Humans and other mammals express five different dopamine receptors, separated into two types. Type 1 dopamine receptors elevate protein kinase A (PKA) signaling and cyclic adenosine monophosphate (cAMP) levels via the alpha subunit G_{α_s} , whereas type 2 receptors inhibit PKA activity via $G_{\alpha_{i/o}}$ thus reducing cAMP levels (Klein et al., 2019; Missale et al., 1998; Tritsch & Sabatini, 2012). Downstream of cAMP and PKA, transcriptional activation mediated by cAMP response element-binding protein leads to the expression of immediate early genes as a response to dopamine signaling (Carlezon et al., 2005; Neves et al., 2002). This pathway is a key regulator of long-term memory (LTM) formation (Alberini & Kandel, 2014; Kaldun & Sprecher, 2019; Kandel, 2012). Moreover, dopamine receptors can modulate internal Ca^{2+} levels by engaging the Phospholipase C signaling pathway. Furthermore, the G-protein β and γ subunits can directly modulate voltage-gated and ligand-gated ion channels (Klein et al., 2019; Ledonne & Mercuri, 2017; Missale et al., 1998; Nishi et al., 2011; Tritsch & Sabatini, 2012). However, the coordination of the receptors and the different dopamine mediated processes in specific circuits remains unexplored. Therefore, an accessible and well-characterized model is required, like the olfactory circuit of *Drosophila melanogaster*. The connectome of this circuit is known, and can be manipulated with the sophisticated genetic tools of the fly (Takemura et al., 2017; Li et al., 2020; Oswald et al., 2015). Similar to mammals, the fly uses dopamine in learning, memory, forgetting, negative and positive reinforcement, locomotion, and sleep and arousal regulation (Berry et al., 2012; Burke et al., 2012; Karam et al., 2020; Sabandal et al., 2021; Sabandal et al., 2020; Siju et al., 2021; Siju et al., 2020; Sitaraman et al., 2015; Tomita et al., 2017; Waddell, 2013; Yamamoto & Seto, 2014). Additionally, L-Dopa, the precursor for dopamine and a powerful receptor agonist, as well as other pharmaceuticals have been shown to function in the fly (Yamamoto & Seto, 2014). Thus, the dopaminergic system can be studied in *Drosophila*, taking advantage of the available genetic tools.

The fly uses four different dopamine receptors. Dop1R1 (Dumb) (Kim et al., 2003; Sugamori et al., 1995) and Dop1R2 (Damb) (Feng et al., 1996; Han et al., 1996) are type 1 receptors. Dop2R is a type 2 receptor (Hearn et al., 2002), and DopEcR is a type 1 dopamine receptor that also uses ecdysone as a ligand (Srivastava et al., 2005). All four receptors were shown to be involved in learning, memory and forgetting (Berry et al., 2012; Karam et al., 2020; Kim et al., 2007; Lark et al., 2017; Qi & Lee, 2014; Qin et al., 2012; Scholz-Kornehl & Schwarzel, 2016; Sun et al., 2020; Zhou et al., 2019). While it is well established that Dop1R1 is crucial for learning and short-term memory (Kim et al., 2007; Qin et al., 2012), the role of the other receptors is less clear. Dop1R2 is an interesting candidate to study, due to its ability to couple to

two different G-proteins, G_{α_s} , which engages the cAMP pathway, and G_{α_q} , which involved in Ca^{2+} -signaling (Han et al., 1996 [\[1\]](#); Himmelreich et al., 2017 [\[2\]](#); Sun et al., 2020 [\[3\]](#)). This could allow Dop1R2 to modulate learning and memory in a complex fashion. The receptor is mainly expressed in the Mushroom body (MB) (Crocker et al., 2016 [\[4\]](#); Croset et al., 2018 [\[5\]](#); Han et al., 1996 [\[1\]](#); Kim et al., 2007 [\[6\]](#); Lark et al., 2017 [\[7\]](#)), an important brain region for olfactory associative learning (Aso, Hattori, et al., 2014; Aso, Sitaraman, et al., 2014; Cognigni et al., 2018 [\[8\]](#)). Previous single cell transcriptomics data show that Dop1R2 is sparsely found in the whole nervous system (in 7.99% of ventral nerve cord cells, and in 13.8% of cells outside the MB) (Allen et al., 2020 [\[9\]](#); Davie et al., 2018 [\[10\]](#)). Work on a mutant line suggests that Dop1R2 is involved in forgetting (Berry et al., 2012 [\[11\]](#)). However, a study using RNAi silencing suggests that the receptor plays a role in memory maintenance (Sun et al., 2020 [\[3\]](#)). As these studies used different learning assays – aversive and appetitive respectively as well as different methods, it is unclear if Dop1R2 has different functions for the different reinforcement stimuli. To resolve this problem, we generated a transgenic line to conditionally knock out *Dop1R2*, in a spatially and temporally specific manner. Using CRISPR-Cas9 and homology-directed repair (HDR), we included FRT sites in the endogenous locus of *Dop1R2* for flippase-mediated excision. In addition, we inserted an HA-Tag, to monitor the spatial localization of the receptor. This also allows us to visualize the efficiency of the flip-out. We used this line to study the role of the receptor in learning and memory in the MB for both aversive and appetitive conditioning. Upon flip-out of the receptor in the MB, 2h memory and long-term memory (LTM) are impaired. Similar results are obtained when we flip out Dop1R2 specifically in the α/β -lobes and α'/β' -lobes of the MB, which are involved in those memory phases. Therefore, Dop1R2 is required in the α/β -lobes and α'/β' -lobes for later memory forms.

Results

Generation of the Dop1R2 conditional knock-out line

To be able to study both the requirement of Dop1R2 in specific neurons as well as the localization of the receptor we generated a transgenic line that allows defined spatial and temporal knock-out of the receptor. In short, we inserted two FRT sites for flippase-mediated excision (Golic & Lindquist, 1989 [\[12\]](#); Gratz et al., 2014 [\[13\]](#)) as well as a 3xHA-tag to study the localization of the receptor. The HA-Tag was chosen to minimize interference with the receptor structure. A representation of the HA-tagged receptor is depicted in **Figure 1A** [\[14\]](#). The 3rd intracellular loop between transmembrane domains (TMDs) five and six as well as the C-terminal tail are required for G-protein binding (Missale et al., 1998 [\[15\]](#)). **Figure 1B** [\[16\]](#) gives an overview of the experimental strategy. The endogenous locus is cut twice by CRISPR-Cas9 to replace the first two coding exons of *Dop1R2* with the corresponding sequence flanked by FRT sites and fused to the HA-Tag, using a donor plasmid as template for homolog-directed repair. The donor plasmid also contains the region directly upstream and downstream of the exchange site as homology arms to align the plasmid. The plasmid carrying the sequence for the two gRNAs as well as the donor plasmid were injected into embryos with maternal Cas9 expression (*nos>Cas9*). Using the Gal4-UAS system to express flippase in neurons of interest the receptor can be irreversibly flipped out in those neurons while keeping it functional in the rest of the brain. The first FRT site was placed in the intron before the first coding exon, which contains all seven TMDs (**Figure 1C** [\[17\]](#)). Since *Dop1R2* has three transcript isoforms with differences in the C-terminus the second FRT site was placed at the end of the last common exon. Thus, FLP-mediated recombination will lead to deletion of the two common coding exons including all TMDs. Successful insertion was verified by sequencing.

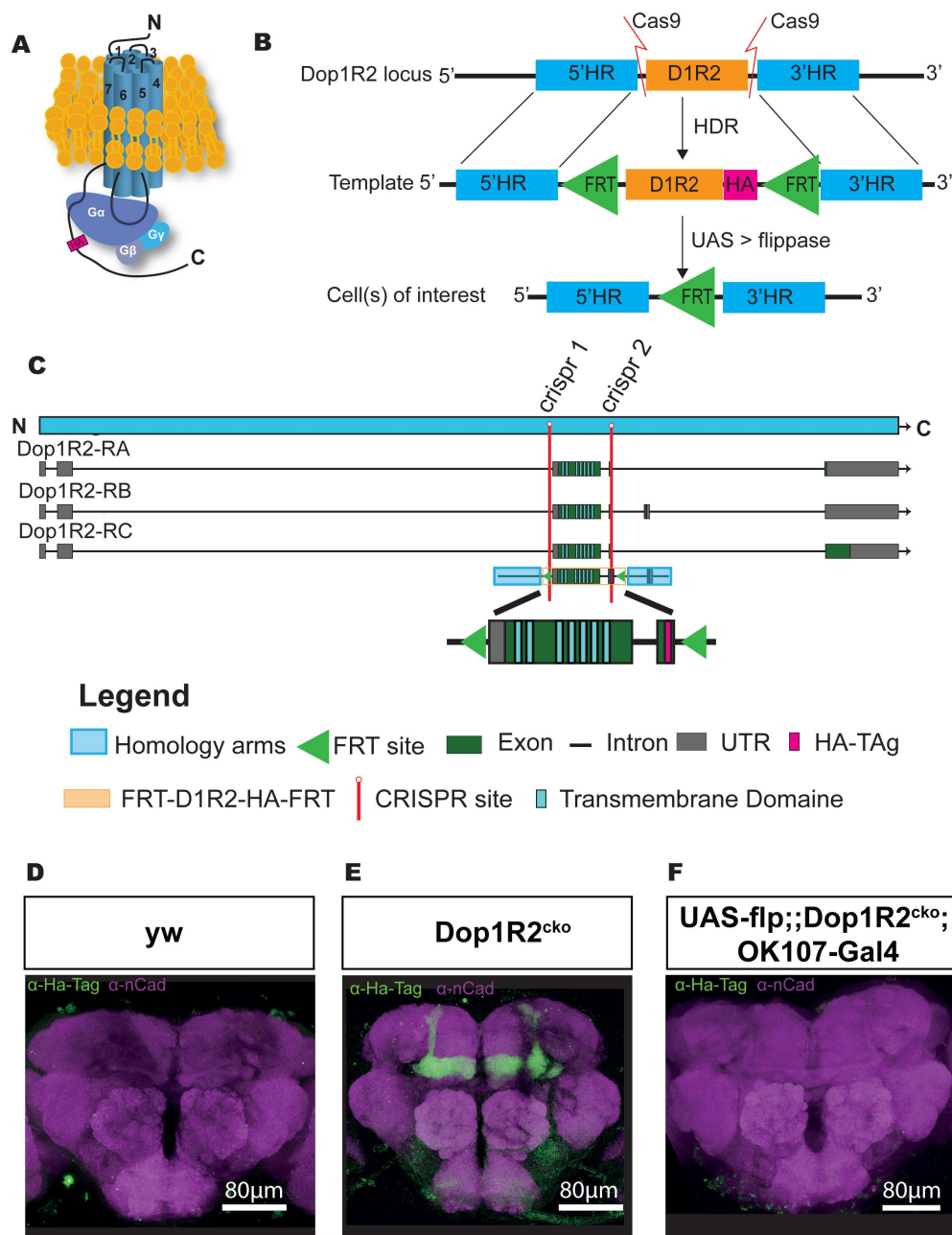


Fig 1.

Generation of a conditional knockout allele for *Dop1R2*.

A) Schematic representation of the receptor structure and interaction with the G-protein complex. The position of the HA-Tag is indicated. B) Schematic representation of the conditional knock-out system. The endogenous *Dop1R2* was replaced using CRISPR-Cas9 mediated homology-directed repair (HDR) from a donor plasmid. The plasmid contained the two common coding exons of *Dop1R2* with an HA-Tag in the C-Terminus and two FRT sites flanking this sequence. In the resulting *Dop1R2^{cko}* allele the inserted HA-Tag and *Dop1R2* sequence can be removed by flippase (FLP) in cells of interest. C) Schematic representation of the *Dop1R2* gene locus with 3 different transcript isoforms. The location of the two used CRISPR sites are highlighted in red. The positions of the transmembrane domains in the isoforms and in the donor plasmid, are indicated. D-F) *Dop1R2::HA* expression in a frontal brain confocal section of D) y, w, E) *Dop1R2^{cko}* or F) *UAS-flp/+;; dop1R2^{cko}*, OK107-Gal4/+ flies aged one week. The HA-Tag was visualized using an anti-HA-Tag antibody (green). Brain structures were labeled with anti-N-cadherin (nCad, magenta) antibody. Scale bar: 80 μ m.

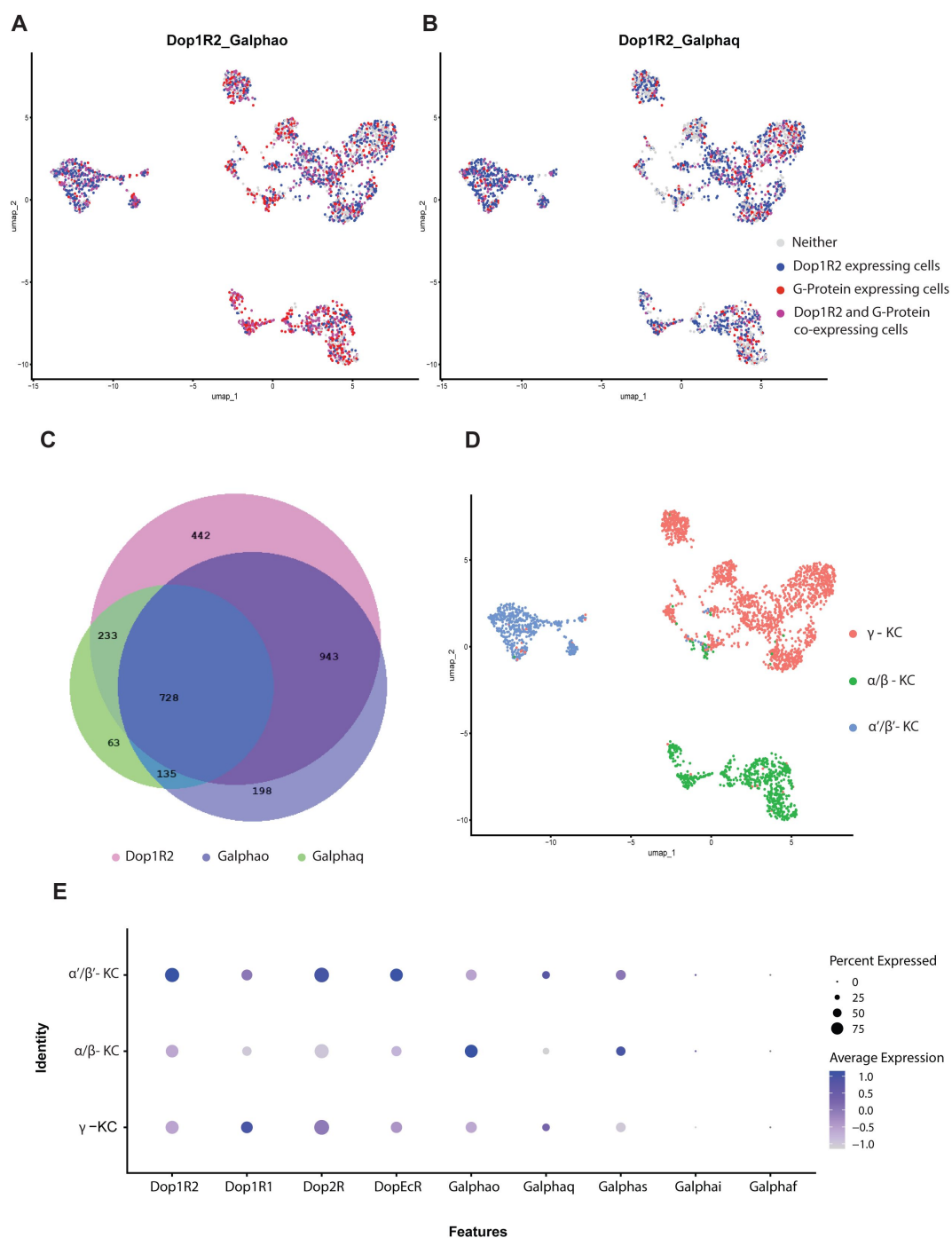


Figure 1 Supplementary 1

Single-cell Transcriptomic analysis of Dop1R2 and various G-proteins in the Mushroom Body neurons.

A-B) Co-expression of Dop1R2 and $G_{\alpha o}$ and $G_{\alpha q}$ respectively. Cells expressing Dop1R2 are depicted in red, while the G-proteins are depicted in blue. While cells expressing both Dop1R2 and one of the G-proteins is shown in magenta. C) Venn Diagram of cells expressing Dop1R2, $G_{\alpha o}$ and $G_{\alpha q}$ in pink, violet and green respectively. The size of the circles corresponds to the number of cells expressing proteins. D) umap clustering of Mushroom body neurons, γ -lobe neurons are shown in pink, α/β -lobe neurons are shown in green while α'/β' -lobe neurons are blue. E) DotPlot of Dopamine receptors and G-Proteins in the different MB lobes. The color and size of the dots represent the average expression and percentage of expression respectively.

Dop1R2^{cko} is expressed in the Mushroom body

To monitor the localization of the tagged dopamine receptors we first stained brains with an antibody against the HA-Tag in one-week-old flies. In the y, w control line ([Figure 1D](#)) the HA-tag does not show any specific staining. In the *Dop1R2^{cko}* line, we see a clear signal in the entire MB ([Figure 1E](#)). This matches previous reports as well as single-cell RNAseq data showing that Dop1R2 is expressed in the MB (Croset et al., 2018; Han et al., 1996).

Flipping the receptor out in the MB using the MB-specific OK107-Gal4 driver in combination with UAS-flp abolishes the signal, demonstrating the efficiency of the FRT sites ([Figure 1F](#)). Therefore, the flip-out system seems to work as planned.

Short-term memory is not affected by the loss of Dop1R2

We wanted to see if flipping out *Dop1R2* in the MB affects memory acquisition and STM by using classical olfactory conditioning. In short, a group of flies is presented with an odor coupled to an electric shock (aversive) or sugar (appetitive) followed by a second odor without stimulus. For assessing their memory, flies can freely choose between the odors either directly after training (STM) or at a later timepoint.

To ensure that the introduced genetic changes to the *Dop1R2* locus do not interfere with behavior we first checked the sensory responses of that line ([Figure 2 S1A-D](#)). The *Dop1R2^{cko}* line shows comparable odor responses ([Figure 2 S1A+B](#)) as well as sugar and shock response ([Figure 2 S1C+D](#)) to the control line. Aversive STM ([Figure 2 S1E](#)), as well as aversive 2h memory ([Figure 2 S1F](#)), are not significantly different from the y, w control line. Moreover, appetitive STM is also comparable to the control ([Figure 2 S1G](#)). Thus, unsurprisingly, the introduced changes to *Dop1R2* do not interfere with normal receptor function.

To test the requirement of Dop1R2 for short-term memory we assessed memory performance directly after training for flies with flipped-out *Dop1R2* in the whole MB or in individual lobes along with the parental controls. The flies were aged for a week before undergoing classical olfactory conditioning. Both aversive STM and appetitive STM were tested. But first, we ensured that the whole MB flip-out line responds normally to the used odors and stimuli. UAS-Flp/+;; *Dop1R2^{cko}*; OK107-Gal4/+ flies show comparable responses as the parental controls to both odors ([Figure 2 S1H+I](#)), shock ([Figure 2 S1J](#)) and sugar ([Figure 2 S1K](#)).

For aversive conditioning, flipping out *Dop1R2* in the whole MB does not change STM compared to parental controls ([Figure 2A](#)). Next, we flipped out *Dop1R2* in the γ -lobe using 5HTR1B-Gal4. This MB lobe is involved in STM (Blum et al., 2009; Trannoy et al., 2011; Zars et al., 2000). Matching the results for the whole MB flip out, we do not see a change in the performance score ([Figure 2B](#)). Similar results are obtained when we flip out the receptor in the α/β -lobes using c739-Gal4 ([Figure 2C](#)) or α'/β' -lobes using c305a-Gal4 ([Figure 2D](#)).

Reward STM is also not changed when *Dop1R2* is flipped out in the whole MB ([Figure 2E](#)) in the γ -lobe ([Figure 2F](#)), the α/β -lobes ([Figure 2G](#)) or the α'/β' -lobes ([Figure 2H](#)). Taken together the results indicate that Dop1R2 is required for STM in none of the MB lobes.

2h memory is impaired by loss of Dop1R2

As Dop1R2 was previously described to be involved in forgetting and/or memory maintenance we wanted to assess later time points after training.

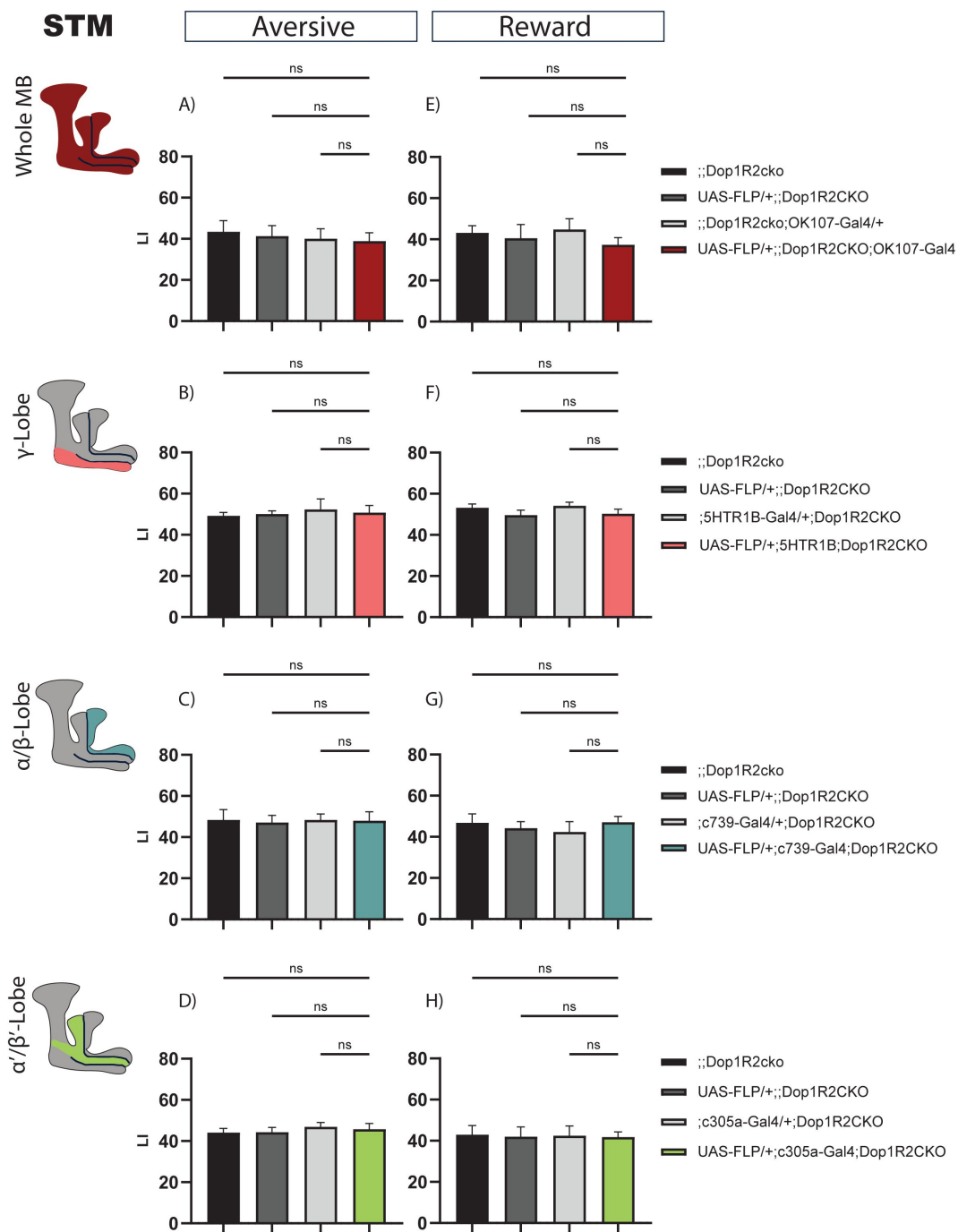


Fig 2.

Short-term memory of flies with knock-out of Dop1R2 in the mushroom Body.

A-D) Aversive training, E-H) reward training. A) and E) Whole MB flip-out using OK107-Gal4 and parental controls. B) and F) γ -lobe flip-out using 5HTR1B-Gal4 and parental controls. C) and G) α/β -lobe flip-out using c739-Gal4 and parental controls. D) and H) α'/β' -lobe flip-out using c305a-Gal4 and parental controls. No performance impairment was observed in any of the tested conditions. See [Fig 2 S1](#) for sensory controls. Bar graphs represent the mean, and error bars represent the standard error of the mean. For each shown graph the N = 12. ns – not significant determined by a One-way ANOVA and Tukey HSD.

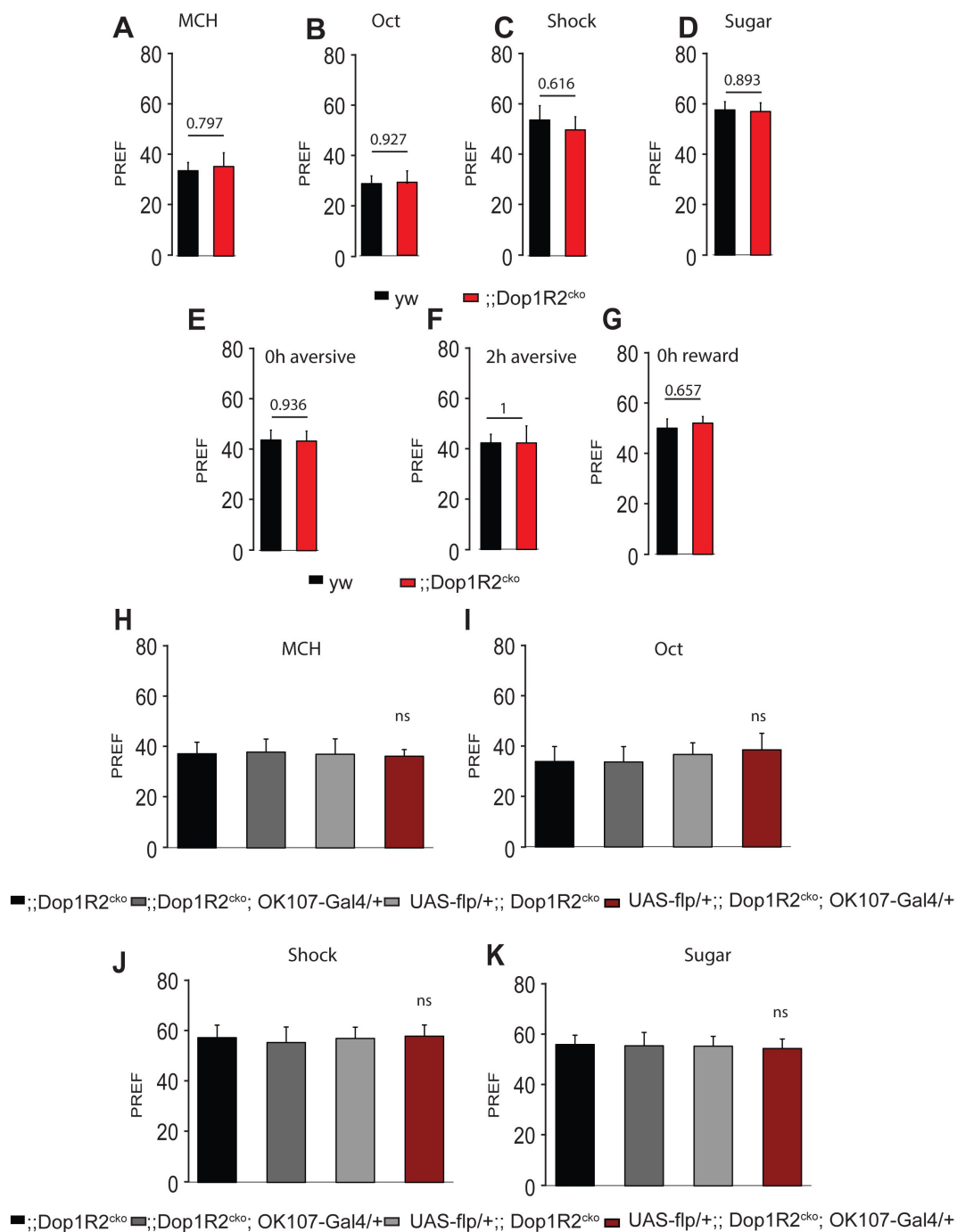


Figure 2 Supplement 1

Sensory tests of the *Dop1R2* conditional knock-out line.

A) Response of *yw* and *Dop1R2^{cko}* to MCH. B) Response of *yw* and *Dop1R2^{cko}* to Oct. C) Response of *yw* and *Dop1R2^{cko}* to Shock. D) Response of *yw* and *Dop1R2^{cko}* to Sugar. E) 0h aversive memory of *yw* and *Dop1R2^{cko}*. F) 2h aversive memory of *yw* and *Dop1R2^{cko}*. G) 0h reward memory of *yw* and *Dop1R2^{cko}*. In all tested conditions the *Dop1R2^{cko}* shows no significant difference to the control line. H-K) Sensory responses of flies with *Dop1R2* knock-out in the whole MB alongside the parental control lines. H) Response to MCH, I) Response to Oct, J) Response to Shock, K) Response to Sugar. Loss of *Dop1R2* in the whole MB does not affect sensory responses. See S1 Table for the data. Bar graphs represent the mean, and error bars represent the standard error of the mean. For each shown graph the N = 12. Asterisks denote significant differences between groups (*p < 0.05, **p < 0.005, ***p < 0.001, ns: not significant) determined by One way ANOVA and Tukey HSD.

First, we looked at two hours after aversive training. Flip-out in the whole Mushroom body leads to a reduced performance score (**Figure 3A** [↗](#)). Next, we asked which MB lobe might cause this reduction. Therefore, we flipped out *Dop1R2* in the γ -lobe using the 5HTR1B-Gal4 driver, characterized in [Aso et al., 2012](#) [↗](#), in the α/β -lobes using c739-Gal4, or α'/β' -lobes using c305a-Gal4. Loss of *Dop1R2* in the γ -lobe does not reduce the memory performance (**Figure 3B** [↗](#)). However, loss of *Dop1R2* in both the α/β -lobes (**Figure 3C** [↗](#)) or the α'/β' -lobes (**Figure 3D** [↗](#)) impaired 2h memory after aversive training. We tested the same for reward memory using sugar as reinforcement. Flip out of *Dop1R2* in the whole MB (**Figure 3E** [↗](#)) results in a reduced performance score. Loss of *Dop1R2* in the γ -lobe does not affect memory performance (**Figure 3F** [↗](#)). As for aversive training, flip out of *Dop1R2* in the α/β -lobes (**Figure 3G** [↗](#)) or the α'/β' -lobes (**Figure 3H** [↗](#)) impaired 2h memory after reward training. However, the reduction is not as severe as for aversive training. Taken together the results indicate that *Dop1R2* is required for aversive 2h memory as well as reward 2h memory in the α/β -lobes and the α'/β' -lobes but is dispensable in the γ -lobe.

24h memory is impaired by loss of *Dop1R2*

Next, we wanted to see if later memory forms are also affected. One cycle of reward training is sufficient to create LTM ([Krashes & Waddell, 2008](#) [↗](#)), while for aversive memory, 5-6 cycles of electroshock-trainings are required to obtain robust long-term memory scores ([Tully et al., 1994](#) [↗](#)). So, we looked at both 24h aversive and appetitive memory. For aversive LTM, the flies were tested on the Y-Maze apparatus as described in [Mohandasan et al., 2022](#) [↗](#).

Flipping out *Dop1R2* in the whole MB causes a reduced 24h memory performance (**Figure 4A**, **E** [↗](#)). No phenotype was observed when *Dop1R2* was flipped out in the γ -lobe (**Figure 4B**, **F** [↗](#)). However, similar to 2h memory, loss of *Dop1R2* in the α/β -lobes (**Figure 4C**, **G** [↗](#)) or the α'/β' -lobes (**Figure 4D**, **H** [↗](#)) causes a reduction in memory performance. Thus, *Dop1R2* seems to be involved in aversive and appetitive LTM in the α/β -lobes and the α'/β' -lobes.

Previous studies have shown mutation in the *Dop1R2* receptor leads to improvement in LTM when a single shock training paradigm is used ([Berry et al., 2012](#) [↗](#)). As we found that it disrupts LTM, we wanted to verify if the absence of *Dop1R2* outside the MB is what leads to an improvement in memory. To that extent, we tested pan-neuronal flip-out of *Dop1R2* flies for 6hr and 24hr memory upon single shock using the *elav*-Gal4 driver. We found that it did not improve memory at both time points (**Figure 4 S1** [↗](#)). Confirming that flipping out *Dop1R2* pan-neuronally does not improve LTM (**Figure 4 S1C** [↗](#)) and highlighting its irrelevance in memory outside the MB.

Developmental defects are ruled out in a temporally restricted *Dop1R2* conditional knockout

To exclude developmental defects in the MB caused by flip-out of *Dop1R2*, we stained fly brains with a FasII antibody. Compared to genetic controls, the anatomical organization as judged by FasII staining in flies lacking *Dop1R2* in the mushroom body was not altered (**Figure 4 S2C** [↗](#)).

To further provide behavioral evidence that the learning defects we observed are not due to developmental knock-out of *Dop1R2*, we generated a *Gal80^{ts}*-containing line, enabling the temporal control of *Dop1R2* knockout in the entire mushroom body (MB). Given that the half-life of the receptor remains unknown, we assessed both aversive short-term memory (STM) and long-term memory (LTM) to determine whether post-eclosion ablation of *Dop1R2* in the MB produced differences compared to our previously tested line, in which *Dop1R2* was constitutively knocked out from fertilization. To achieve this, flies were maintained at 18°C until eclosion and subsequently shifted to 30°C for five to seven days. On the fifth day, training was conducted, followed by memory testing. Our results indicate that aversive STM was not significantly impaired in *Dop1R2*-deficient MBs compared to control flies (**Figure 4 S3** [↗](#)), consistent with our previous

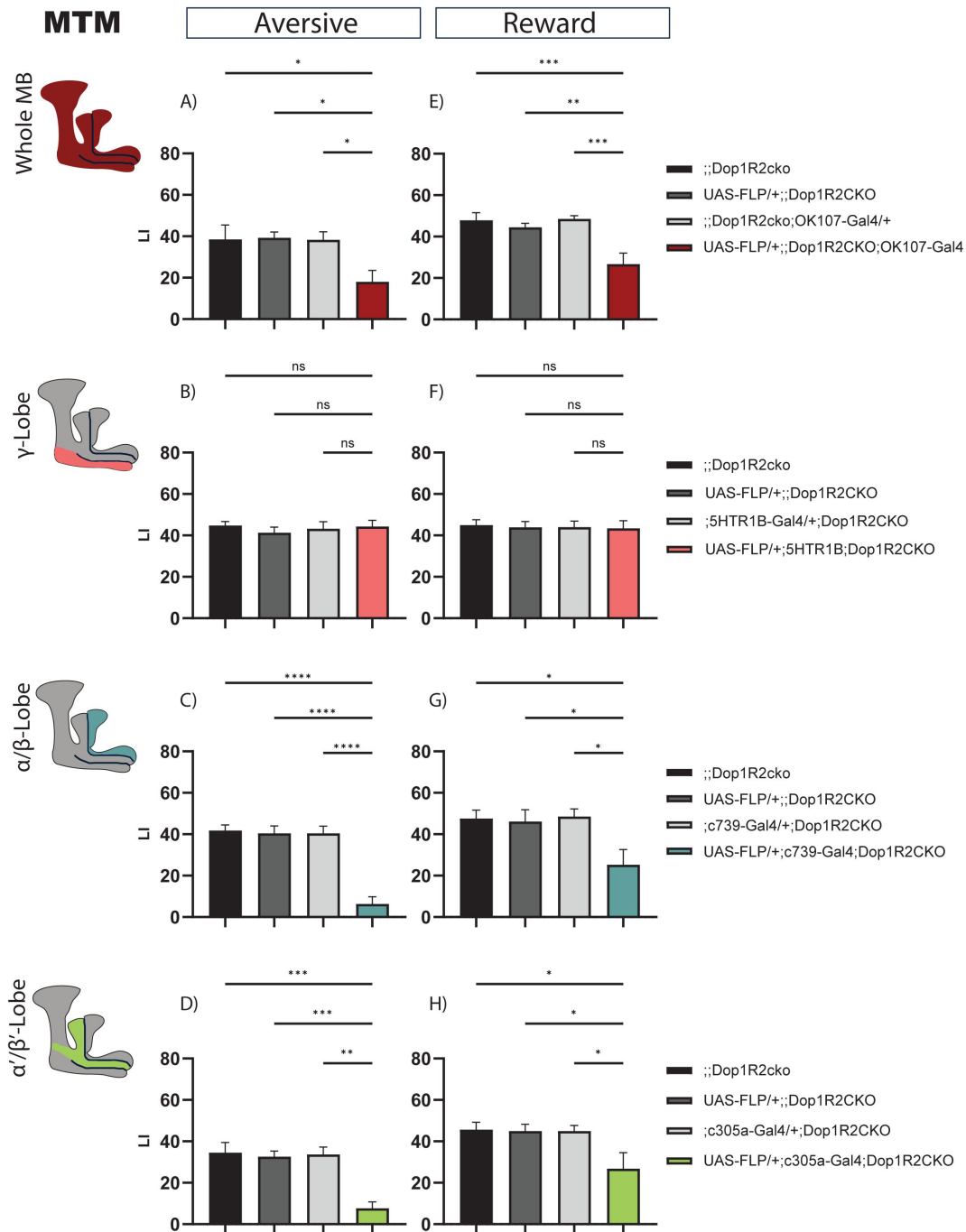


Fig 3.

2h memory of flies with knock-out of Dop1R2 in the mushroom Body.

A-D) Aversive training, E-H) reward training. A) and E) Whole MB flip-out using OK107-Gal4 and parental controls. B) and F) γ -lobe flip-out using 5HTR1B-Gal4 and parental controls. C) and G) α/β -lobe flip-out using c739-Gal4 and parental controls. D) and H) α'/β' -lobe flip-out using c305a-Gal4 and parental controls. For whole MB flip-out, α/β -lobes and α'/β' -lobes both aversive and appetitive 2h memory performance is impaired. Loss of Dop1R2 in the γ -lobe does not affect 2h memory. See **Fig 2 S1** for sensory controls. Bar graphs represent the mean, and error bars represent the standard error of the mean. For each shown graph the N = 12. Asterisks denote significant differences between groups (* p < 0.05, ** p < 0.005, *** p < 0.001, **** p < 0.0001, ns: not significant) determined by One way ANOVA and Tukey HSD (Panels A-C, E-H) and Kruskal-Wallis with Dunn's multiple comparisons test due to non-normal distribution (Panel D).

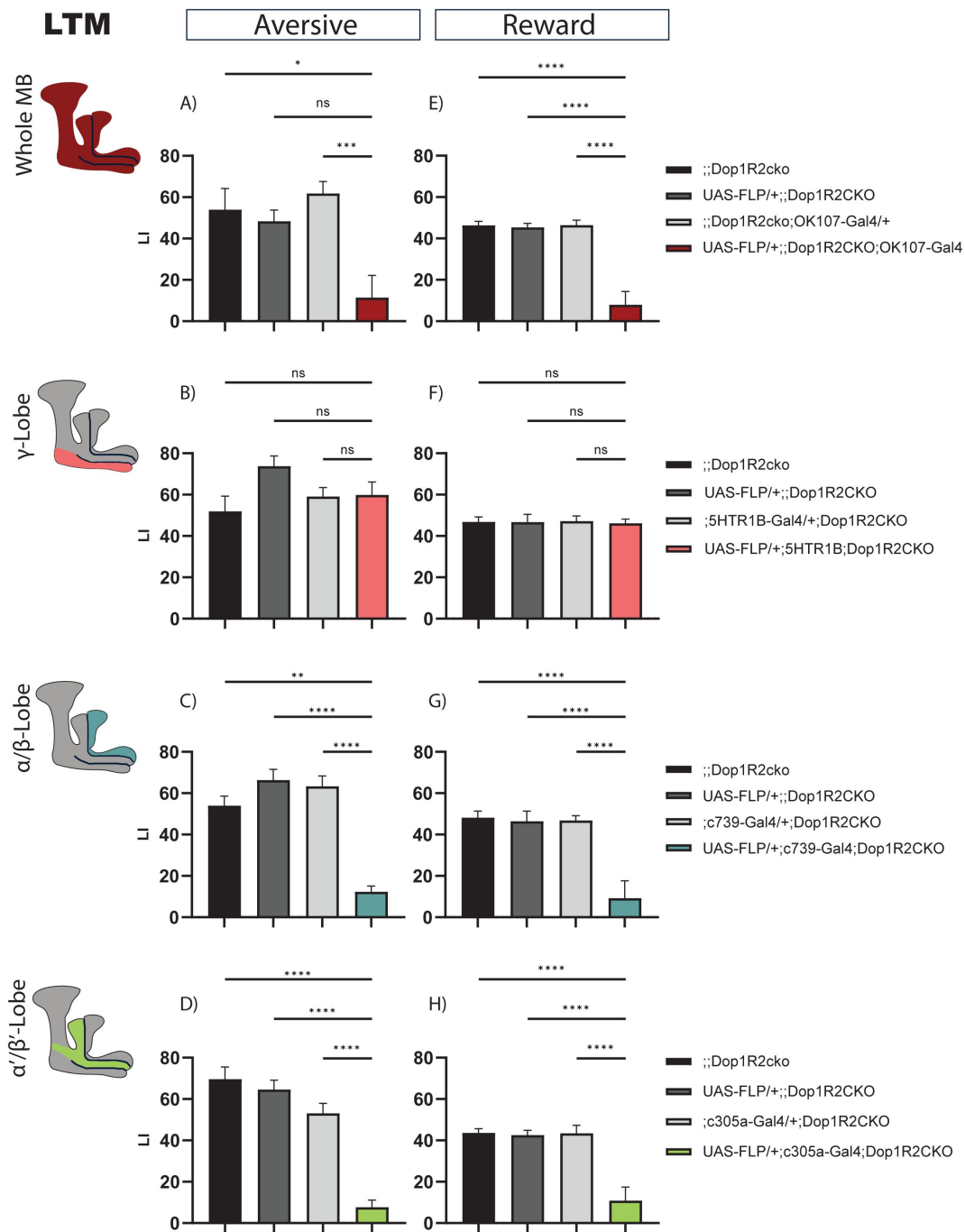


Fig 4.

24h memory of flies with knock-out of Dop1R2 in the mushroom Body.

A-D) Aversive training, E-H) reward training. A) and E) Whole MB flip-out using OK107- Gal4 and parental controls. B) and F) γ -lobe flip-out using 5HTR1B-Gal4 and parental controls. C) and G) α/β -lobe flip-out using c739-Gal4 and parental controls. D) and H) α'/β' -lobe flip-out using c305a-Gal4 and parental controls. For whole MB flip-out, α/β -lobes and α'/β' -lobes aversive and appetitive 24h memory performance is impaired. Loss of Dop1R2 in the γ -lobe does not affect 24h memory. See **Fig 2 S1** for sensory controls. Bar graphs represent the mean, and error bars represent the standard error of the mean. For each shown graph included in the reward training experiment the N = 12, while for the graphs included in the aversive training experiment the N = 14. Asterisks denote significant differences between groups (*p < 0.05, **p < 0.005, ***p < 0.001, ****p < 0.0001, ns: not significant determined by Kruskal-Wallis with Dunn's multiple comparisons test due to non-normal distribution (Panels A-C) and One way ANOVA and Tukey HSD (Panels D-H).

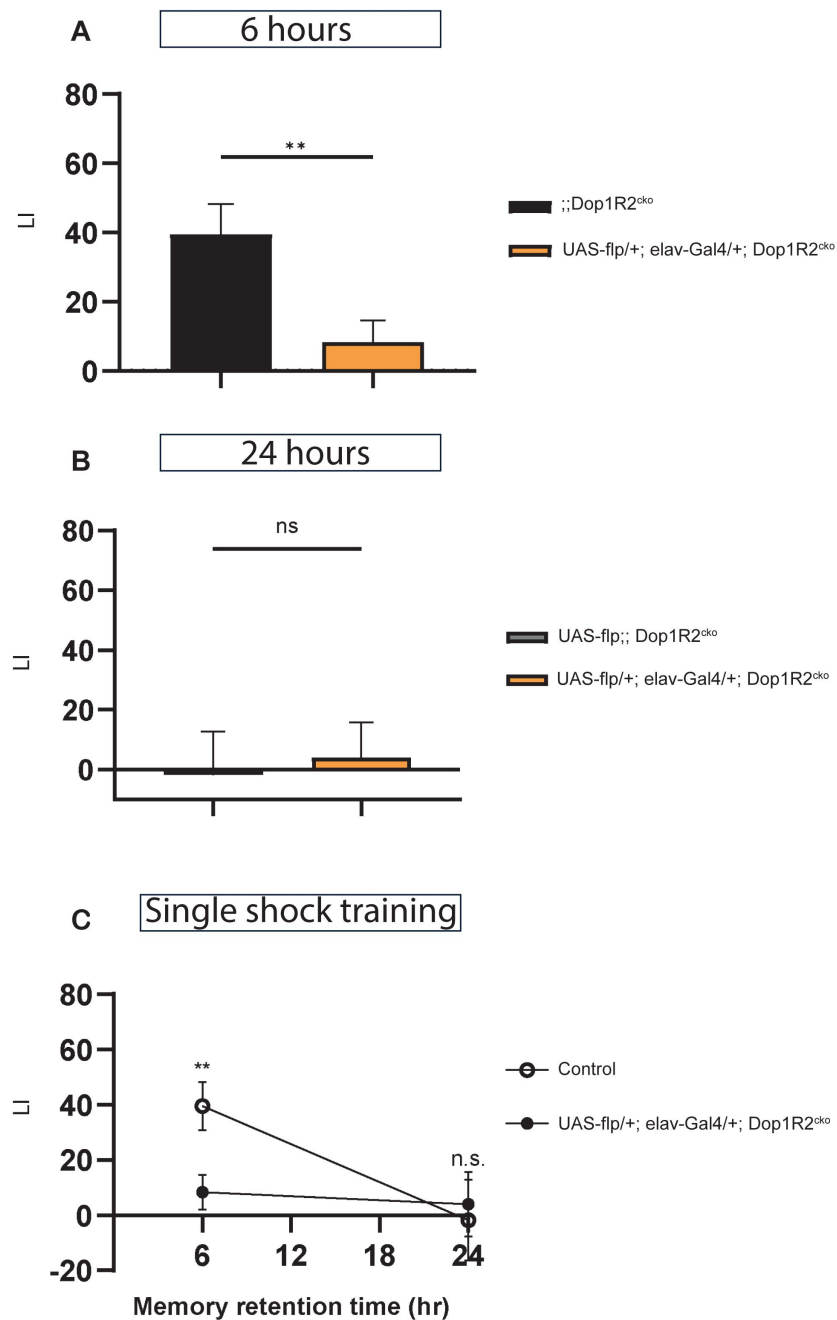


Figure 4 Supplement 1.

Memory retention upon single shock aversive training in *Dop1R2* pan-neuronal knockout flies.

A) 6 hours memory performance of *Dop1R2* pan-neuronal knockout flies using elav-Gal4. B) 24 hours memory performance of *Dop1R2* pan-neuronal knockout flies using elav-Gal4. C) Memory retention over 24 hours in *Dop1R2* pan-neuronal knockout flies. Lack of the receptor in the whole nervous system does not improve memory retention suggesting that *Dop1R2* sparsely expressed outside of the MB might not be regulating any memory process. Single shock training is not sufficient to retain aversive memory for 24 hours in control flies either. For each graph shown the N = 14. Asterisks denote significant differences between groups (*p < 0.05, **p < 0.005, ***p < 0.001, ns: p > 0.05) determined by unpaired t-test. The one-sample t-test showed no significant difference from zero for all the groups apart from the control line in the 6 hours memory experiment.

Figure 4 Supplement 2.

Gamma lobe development in the *Dop1R2^{cko}* carrying flies.

A-C) FasII expression in a frontal view of whole brain immunostaining, highlighting the Mushroom body. The *Dop1R2^{cko}* modification does not affect gamma lobe (white arrows) development. Images were taken at 63x magnification, Scale bar: 50 μ m.

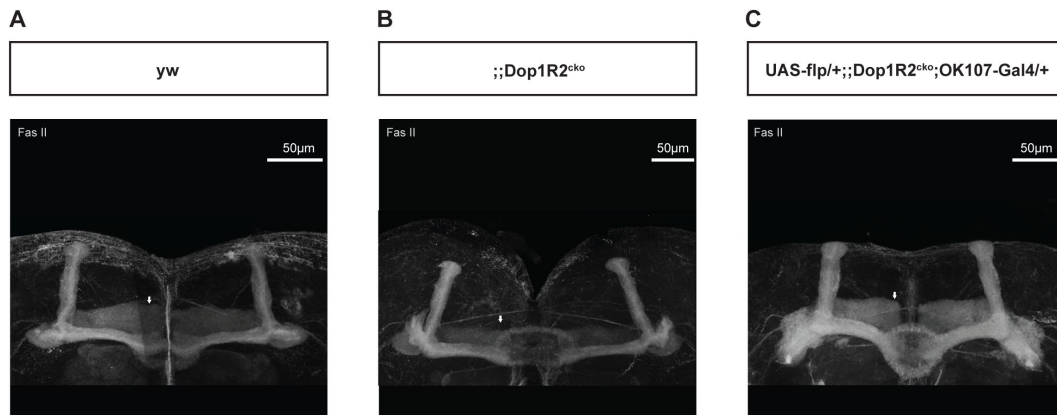
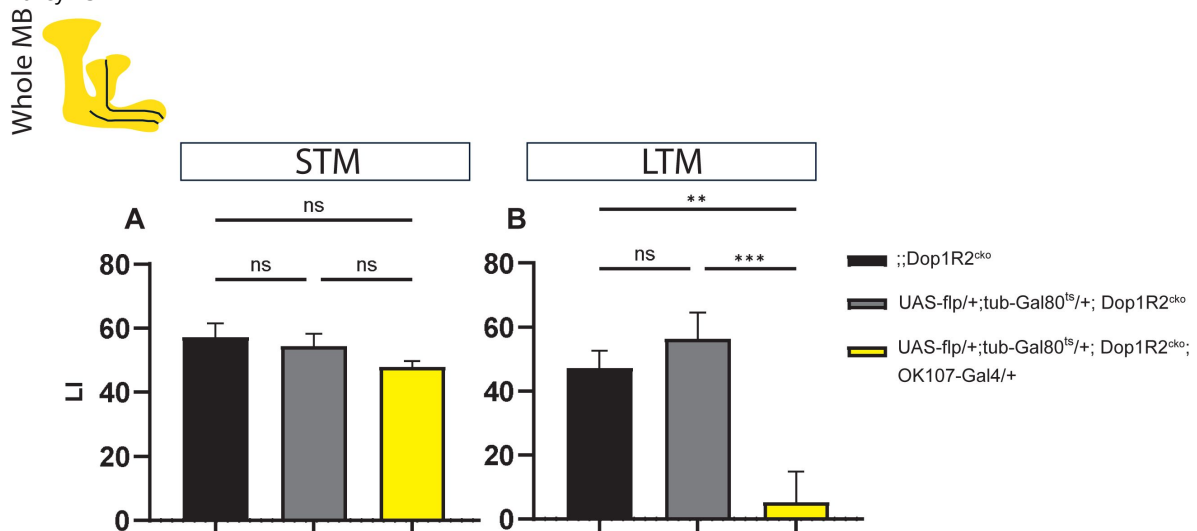


Figure 4 Supplement 3.

A temporally restricted knock-out of *Dop1R2* in the mushroom Body rules out developmental defects.

A) and B) Aversive training of flies with whole Mushroom Body flip-out at 0 days post-eclosion, using the flies *;;Dop1R2^{cko}* and *UAS-flp/+; tub-Gal80^{ts}/+; Dop1R2^{cko}* as control and *UAS-flp/+; tub-Gal80^{ts}/+; Dop1R2^{cko}; OK107-Gal4/+* the experimental, temporally restricted knockout flies. A) Short term memory tested 0 hours after training. B) Long term memory tested 24 hours after training. Loss of *Dop1R2* in the whole mushroom body exclusively after eclosion had no effect on STM but significantly impaired memory retention at 24 hours. Bar graphs represent the mean, error bars represent the standard error of the mean. For the STM graph the N = 9, while for the LTM graph the N = 14. Asterisks denote significant differences between groups (*p < 0.05, **p < 0.005, ***p < 0.001, ****p < 0.0001, ns: not significant) determined by One way ANOVA and Tukey HSD



findings (Figure 2). However, aversive LTM was significantly impaired relative to control lines (Figure 4 S3), which also aligned with prior observations. These findings strongly indicate that memory loss caused by Dop1R2 flip-out is not due to developmental defects.

Discussion

We have generated a conditional knock-out line for the dopamine receptor Dop1R2 following a similar approach as in Widmer et al., 2018. To achieve this, FRT sites were inserted in front of the start codon and in the C-Terminus of Dop1R2 using CRISPR-Cas9 mediated homology-directed repair. In addition, an HA-Tag was inserted to be able to visualize the dopamine receptor expression as well as verify successful flip out. Using an anti-HA-Tag antibody we were able to visualize Dop1R2 in the MB of the generated line. This matches previous reports for the expression of *Dop1R2* (Crocker et al., 2016; Croset et al., 2018; Han et al., 1996; Kim et al., 2007; Sun et al., 2020). Moreover, upon knocking out *Dop1R2* with an MB-specific driver, the HA-tag labeling disappears indicating that the conditional knock-out system works. The HA-Tag could also be useful to study the subcellular localization of Dop1R2 within the MB lobes, for example if it is close to synapses of dopaminergic neurons.

To get a better overview of Dop1R2's role in the Mushroom body we analyzed appetitive memory at different timepoints after training in the individual MB lobes. Loss of Dop1R2 in the whole MB as well as the α/β -lobe or the α'/β' -lobe impairs 24h reward memory. This observation matches previous studies. Using *Dop1R2-RNAi* in the MB, Sun et al., 2020 showed that STM is intact, while LTM is impaired. When knocking down the Raf/MAPK pathway they get a similar phenotype. Moreover, expression of a constitutive active *Raf* allele rescues the Dop1R2 dependent LTM deficit and Dop1R2 seems to be required for the phosphorylation of the MAPK. Therefore, Dop1R2 dependent activation of the Raf/MAPK pathway is required for stabilization of reward LTM memory. Another study shows that reward LTM requires food with a high energetic value (Musso et al., 2015). This signal seems to be relayed by the dopaminergic neuron (DAN) MB-MP1 (PPL1- γ 1pedc), that showed temporally restricted oscillating activity early post-training (Musso et al., 2015; Pavlowsky et al., 2018). Furthermore, knock-down of Dop1R2 using RNAi impaired reward LTM while leaving STM intact. Therefore, dopaminergic signaling through MB-MP1 and Dop1R2 could indicate the energetic value of the reward and decide if LTM should be formed or not. Interestingly, the MB-MP1 DAN arborizes in the spur of the γ -lobe as well as the inner core of the pedunculus, which consists of the axons of the α/β Kenyon Cells (KCs) (Tanaka et al., 2008), which according to our results require Dop1R2 for reward LTM. Loss of Dop1R2 in the Mushroom Body Output neuron MVP2 (MBON- γ 1pedc α/β) also impaired reward LTM while leaving STM intact (Pavlowsky et al., 2018). This GABAergic MVP2 MBON forms a feedback circuit with the MB-MP1 DAN. After training, the oscillatory activity of MB-MP1 is enhanced, while MVP2 is inhibited. After 30min, MVP2 gets activated and MB-MP1 is inhibited. Moreover, Dop1R2 seems to be required for modulating this feedback loop.

The MBONs are an important contributor to memory. Moreover, MBONs receive dopaminergic input and seem to express dopamine receptors (Crocker et al., 2016). We showed that Dop1R2 is required in the α/β -lobe and the α'/β' -lobe for reward LTM formation. Interestingly, MBONs which are involved in reward LTM (Ichinose et al., 2015; Oswald et al., 2015; Pla  ais et al., 2013) have arborization in the α/β -lobe and the α'/β' -lobe. Thus, Dop1R2 could modulate these connections as well.

In all, Dop1R2 is required for reward LTM formation and loss of the receptor impairs LTM. Dop1R2 seems to influence reward LTM in different ways. Firstly, by acting on the Raf-MAPK pathway to stabilize memory, secondly by relaying the energetic content of the reward and third, by modulating the MB-MP1-MVP2 loop.

For aversive conditioning, we observe that loss of Dop1R2 in the MB leads to impaired 2h memory and LTM, whereas STM is intact. Moreover, Dop1R2 seems to be required in the α/β -lobe and the α'/β' -lobe. A previous study using a mutant for *Dop1R2*, that also affects the neighboring gene *GC1907*, observed higher memory retention (Berry et al., 2012). Further, lack of Dop1R2 impairs reversal learning. It is proposed that Dop1R2 acts on the RAC-forgetting pathway (Cervantes-Sandoval et al., 2016; Shuai et al., 2010). Interestingly, Dop1R2 can act on two different downstream second messenger systems by using two different G-proteins (Himmelreich et al., 2017). By coupling to $G_{\alpha s}$ the cAMP pathway is activated. By coupling to $G_{\alpha q}$ the Ca^{2+} messenger system is activated. Furthermore, knocking down $G_{\alpha q}$ pan-neuronally or in the MB leads to memory enhancement 3h after training, but not 6h after training (Himmelreich et al., 2017). Thus, Dop1R2 could regulate the RAC forgetting pathway through $G_{\alpha q}$. Some RNA binding proteins and Immediate early genes help maintain identities of Mushroom body cells and are regulators of local transcription and translation (de Queiroz et al., 2025; Raun et al., 2025). So, the availability of different G-proteins may change in different lobes and during different phases of memory. The G-protein via which GPCRs signal may depend on the pool of available G-proteins in the cell/sub-cellular region (Hermans, 2003). Therefore, Dop1R2 may signal via different G-proteins in different compartments of the Mushroom body and also different compartments of the neuron. We looked at $G_{\alpha o}$ and $G_{\alpha q}$ as they are known to have roles in learning and forgetting (Ferris et al., 2006; Himmelreich et al., 2017). We found that Dop1R2 co-expresses more frequently with $G_{\alpha o}$ than with $G_{\alpha q}$ (Figure 1 S1). While there is evidence for Dop1R2 to act via $G_{\alpha q}$ (Himmelreich et al., 2017). It is difficult to determine whether this interaction is exclusive, or if Dop1R2 can also be coupled to other G-proteins. It will be interesting to determine the breadth of G-proteins that are involved in Dop1R2 signaling.

Berry and colleagues also observed that blocking output of DANs inhibits forgetting, while activating DANs accelerates forgetting (Berry et al., 2012). This modulation seems to require ongoing activity of the MP1 DAN together with further DANs. Interestingly, using a spaced training protocol and a *Dop1R2* RNAi knockdown, another study showed impaired LTM (Plaçais et al., 2017). Furthermore, after spaced training, flies have a higher energy uptake and the energy metabolism is upregulated in the MB. This increase in energy consumption is mediated by dopaminergic signaling from the MB-MP1 DAN. Loss of Dop1R2 in the MB abolishes the increase in energy consumption. Therefore, Dop1R2 seems to be important for aversive LTM formation by regulating energy consumption. Thus, like for reward LTM formation, aversive LTM seems to require sufficient energy. Starved flies reduce the formation of aversive LTM (Plaçais & Preat, 2013; Plaçais et al., 2012). This information seems to be relayed through ongoing oscillation of MB-MP1 and Dop1R2 after training. In addition, the MVP2 MBON might also be involved (Ueoka et al., 2017). Therefore, the gating mechanism in both aversive and reward LTM formation seems to require Dop1R2 (Pavlovsky et al., 2018).

In addition, the MAPK signaling pathway might also be required in aversive LTM formation by activating transcription factors like CREB and c-fos (Miyashita et al., 2018).

Both Dop1R2 and the ongoing activity of MB-MP1 seem to have multiple roles directly after training in a short time window (Berry et al., 2012; Plaçais et al., 2017; Plaçais et al., 2012). The circuit acts like a gating mechanism for LTM to ensure, that there is enough energy for continuing LTM formation. The MBON MVP2 acts as feedback loop to regulate the activity of the DAN. Moreover, both the RAC and the Raf/MAPK signaling pathway seem to be engaged to either forget or stabilize the memory. In aversive memory formation, loss of Dop1R2 could lead to enhanced or impaired memory, depending on the activated signaling pathways. The signaling pathway that is activated further depends on the available pool of secondary messengers in the cell (Hermans, 2003) which may be regulated by the internal state of the animal. However, it remains unclear, how all of these aspects are integrated and if there is a hierarchical order.

DANs can produce aversive and appetitive associations depending on the temporal presentation of odor cue and reinforcement stimulus (Handler et al., 2019 [DOI](#)). Thus, dopaminergic signaling can modify the KC-MBON synapses bi-directionally. The Dop1R1-G_{as}-cAMP pathway seems to detect the temporal coincidence of the stimuli whereas the Dop1R2-G_{αq}-Ca²⁺ pathway detects the temporal ordering (Handler et al., 2019 [DOI](#)). Dop1R2 mutant flies seem to be able to form an odor association but are not able to update it. Further, Dop1R2 is required for the potentiation of the KC-MBON synapse following backward pairing.

Taken together, Dop1R2 has multiple roles during memory formation and integrates different signals, including the detection of the order of stimuli, the internal state, like energy levels and forgetting and maintenance signals.

The receptor does not seem to be required for STM but for later timepoints. Previous studies looking at the temporal requirement of the lobes (Güven-Ozkan & Davis, 2014 [DOI](#); Perisse et al., 2013 [DOI](#)) defined the γ -lobe to be responsible for memory acquisition (Blum et al., 2009 [DOI](#); Trannoy et al., 2011 [DOI](#); Zars et al., 2000 [DOI](#)). The α/β -lobe and its output is involved in LTM (Akmal et al., 2011 [DOI](#); Blum et al., 2009 [DOI](#); Cervantes-Sandoval et al., 2013 [DOI](#); Huang et al., 2012 [DOI](#); Ichinose et al., 2015 [DOI](#); Krashes & Waddell, 2008 [DOI](#); Trannoy et al., 2011 [DOI](#)). The function of the α'/β' -lobe seems to be LTM consolidation (Cervantes-Sandoval et al., 2013 [DOI](#); Krashes & Waddell, 2008 [DOI](#)). However, both lobes also seem to have a role in middle-term memory (Bouzaiane et al., 2015 [DOI](#); Scheunemann et al., 2012 [DOI](#); Shyu et al., 2019 [DOI](#); Turrel et al., 2022 [DOI](#)). As loss of Dop1R2 in the γ -lobe or the whole MB does not impair STM, we conclude that the receptor is not required for this memory type. However, *Dop1R2* is expressed in the γ -lobe (Crocker et al., 2016 [DOI](#); Croset et al., 2018 [DOI](#)), so it might be required for other types of behaviors.

The impairment of reward LTM upon knock-out of *Dop1R2* in the α/β -lobe and the α'/β' -lobe matches the described role of these neurons. So, both the α/β -lobe and the α'/β' -lobe require Dop1R2 for LTM. Interestingly, both lobes also seem to require Dop1R2 for two-hour memories. As the MB-MP1 DAN is active in this time window as well, this would suggest that Dop1R2 function at this time point is important for correct LTM formation.

This would indicate, that Dop1R1 is the main contributor to STM while Dop1R2 is responsible for later memory stages. It would be interesting to know how the switch from Dop1R1 dependency to Dop1R2 occurs.

As Dop1R2 is required in the MBON MVP2 for reward LTM, it would be exciting to see if it is also required in other MBONs or neurons outside the MB. Besides learning and memory, the MB also uses dopamine to regulate sleep. This tool offers the opportunity to study both aspects in neurons of interest.

The genetic tool we generated here to study the role of the Dop1R2 dopamine receptor in cells of interest, is not only a good substitute for RNAi knockouts, which are known to be less efficient in insects and in many instances may cause a partial, hypomorphic phenotype (Joga et al., 2016 [DOI](#)), but also provides versatile possibilities as it can be used in combination with the powerful genetic tools of *Drosophila*.

Using this line, we could show, that Dop1R2 is specifically required for later memory stages of both aversive and appetitive memory in the α/β -lobe and the α'/β' -lobe.

Methods

Lead Contact and Data Availability

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Simon Sprecher (simon.sprecher@unifr.ch). All used data is in this manuscript.

Fly Husbandry

Drosophila melanogaster flies were reared in plastic vials on standard cornmeal food (12g Agar, 40g Sugar, 40g Yeast, 80g Cornmeal per Liter) and transferred to fresh food vials every 2-3 days. Flies were generally kept at 25°C, 60-65% humidity, and exposed to 12 h light and 12 h darkness with light onset at 8 am. The following fly lines were used: *y[1] w[*] P{y[+t7.7]=nos-phiC31|int.NLS}X; P{y[+t7.7]=CaryIP}su(Hw)attP6* (abbreviated *nos>Cas9* in this paper; BL 32232) for microinjection and as PCR template, *w*; *Dr e/TM3* (BL 36305) was used as 3rd chromosomal balancer line. *w[1118], 20XUAS-FLPG5.PEST* (BL 55807) was used as *UAS-flp*. OK107-Gal4 (106098) was obtained from the Kyoto stock center. The 5HTR1B-Gal4 line (BL 27636) and the c305a-Gal4 (BL 30829) line are from Bloomington stock center. The c739- Gal4 line was gifted to us by H. Tanimoto (Tohoku University Japan). The *y w* (BL 1495) line was used as the control line.

UAS-flp;;Dop1R2^{cko} flies and *Gal4;Dop1R2^{cko}* flies were crossed back with *;;Dop^{cko}* flies to obtain appropriate genetic controls which were heterozygous for *UAS* and *Gal4* but not *Dop1R2^{cko}* to control for genetic background.

Generation of *Dop1R2^{cko}*

For generating the conditional knock-out line we needed three regions: 1) The 5' flanking sequence upstream of the first FRT site as homologous region, 2) The 3' flanking sequence downstream of the second FRT site as homologous region, 3) the sequence in between the two FRT sites hereafter named *Dop1R2* coding fragment. To obtain the dopamine receptor fragments, genomic DNA from *nos>Cas9* flies was used as the template for the PCRs. The primers used for the different PCR fragments are shown in **Table 2**. The 3' and 5' fragments of *Dop1R2* were sub-cloned into *pBluescript II SK(+)* vector (*pBS*) with adequate restriction enzymes – *SpeI* and *SmaI* for the 5' fragment and *XhoI* and *Acc65I* for the 3' fragment. The *Dop1R2* coding fragment was cloned with the Invitrogen TOPO kit. After sequence confirmation the fragments were cloned subsequently into *pBS-FRT-3xHA-FRT*. The vector is modified from the vector used by Widmer et al. (Widmer et al., 2018). The GFP was replaced by 3xHA using 3xHA *Nco*

Fw: (cacatggTACCCATACGATGTTCTGACTATGCGGGCTATCCCTATGACGTCCCG
GACTATGCAGGATCCTATCCATATGACGTTCCAGATTACGCTgca) and 3xHA H3

Rev: (agcttgCAGCGTAATCTGGAACGTCATATGGATAGGATCCTGCATAGTCCGGGAC
GTCATAGGGATAGCCGCATAGTCAGGAACATCGTATGGGTac) as oligos whereas the FRT site and the *pBS* backbone were kept. The *Dop1R2* fragment was cloned in-frame in front of the 3xHA tag using *AgeI* and *BstEII*-HF as restriction enzymes.

The two guide RNAs were chosen using crisper optimal target finder (<http://targetfinder.flycrispr.neuro.brown.edu/>). The used sequences (**Table 2**) were integrated into PCR primers to amplify a fragment of the *pCFD4-U6_1_6_3tandemgRNAs* plasmid (a gift from Simon Bullock, addgene plasmid # 49411 (Port et al., 2014)) that was cloned into *BbsI* digested *pCFD4-U6_1_6_3tandemgRNAs* vector using Gibson assembly (NEB). Insertion of the gRNAs was confirmed by sequencing. A mix containing 0.4 mg/ml template and 0.2 mg/ml of the guide RNAs plasmid was injected into *nos>Cas9* embryos. The injected flies were crossed with a 3rd

Key Resources Table				
Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	y[1] w[*] P{y[+t7.7]=nos-phiC31\int.NLS}X; P{y[+t7.7]=CaryIP}su(Hw)attP6	Bloomington Stock Center	RRID:BDSC_3232	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	w;; Dr e/TM3	Bloomington Stock Center	RRID:BDSC_36305	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	w[1118], 20XUAS-FLPG5.PEST	Bloomington Stock Center	RRID:BDSC_55807	
Genetic Reagent <i>Drosophila</i>	OK107-Gal4	Kyoto Stock Center	DGGR_106098	

Table 1

Key Resource Table

<i>melanogaster</i> both male and female				
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	5HTR1B-Gal4	Bloomington Stock Center	RRID:BDSC_27636	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	c739-Gal4	Hiromu Tanimoto (Tohoku University Japan)	RRID:BDSC_7362	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	c305a-Gal4	Bloomington Stock Center	RRID:BDSC_30829	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	y w	Bloomington Stock Center	RRID:BDSC_1495	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	FRT-Dop1R2-HA-FRT Dop1R2 ^{cko}	This paper		See Materials and Methods section: Creation of Dop1R2 ^{cko}

Table 1 (continued)

antibody	Mouse monoclonal α -HA clone 12CA5	Roche	RRID:AB_514505 11583816001	1:200
antibody	Mouse monoclonal anti-Fasciclin II	Developmental Studies Hybridoma Bank	RRID:AB_528235	1:50
antibody	Rat monoclonal α -Drosophila-N-cadherin (Ncad)	Developmental Studies Hybridoma Bank	RRID:AB_528121 DN-Ex #8	1:30
Plasmid	pCFD4-U6_1_6_3tandemgRNAs plasmid	Simon Bullock	RRID:Addgene_49411	

Table 1 (continued)

chromosomal balancer line. The F1 generation was crossed again with the 3rd chromosomal balancer flies. As soon as eggs or larvae were visible the adults were sacrificed to check for the *Dop1R2* construct via PCR. Positive hits were then sequenced, and a stock was established.

Immunohistochemistry

Brains from 5–8-day old flies were dissected in PBS (Bio-Froxx 1346LT050) and fixed in 3.7% formaldehyde for 25 min at RT. The brains were washed with 1X PBS containing 0.5% Triton X-100 (Carl Roth 3051.3) (0.5% PBST) before incubating the primary antibodies o/n at 4°C. The primary antibodies were mouse α -HA clone 12CA5 (Roche) 1:200, rat α -Drosophila-cadherin (Ncad) (Iowa H.B: DN-EX8) 1:30 or mouse α -Fasciclin II (FasII) (Iowa H.B: 1D4). After the brains were washed in 0.5% PBST again, they were incubated o/n at 4°C with the secondary antibodies. The following secondary antibodies were used: Goat α -rat Alexa 647 (Molecular probes A-21247) and Goat α -Mouse Alexa 488 (Molecular probes A11029) 1:200. The brains were washed again and mounted in self-made mounting media (90% Glycerol (Fischer Scientific Catalog No. BP229-1), 0.5% N-propyl gallate (Sigma P3130), 20 mM Tris (Fischer Scientific, Catalog No. BP152-5), pH 8.0) (Adapted from NIC Harvard Medical School). The brains were imaged using a confocal microscope (Leica STELLARIS 8 FALCON) at 40X magnification with the Plan APO 40x/1.10 water immersion objective at 1,024 × 1,024 pixels resolution and 600Hz scan rate. Images were processed with Fiji ImageJ and Adobe Illustrator.

Learning Apparatus

For behavior experiments, we used a memory apparatus that is based on Tully and Quinn's design and modified to allow conducting four memory experiments in parallel (CON-Elektronik, Greussenheim, Germany). Experiments were performed at 23–25°C and 65–75% relative humidity. The training was performed in dim red light and memory tests were done in complete darkness. The two odors used were 3-octanol (3-Oct) (Sigma-Aldrich 218405) and 4-methyl-cyclohexanol (MCH) (Sigma-Aldrich 66360) diluted in paraffin oil (Sigma-Aldrich 18512) 1:100 respectively. 260 μ l of the diluted odors were presented in a plastic cup of 14 mm in diameter. A vacuum membrane pump ensured odor delivery at a flow rate of 7 l/min.

Aversive olfactory conditioning

For aversive conditioning, groups of 50–100 flies of mixed sex were loaded in tubes lined with an electrifiable copper grid. The position in the machine and the sequence in which the genotypes were tested were randomized. Experiments in which more than half of the flies died, the flies did not move or there were technical problems with the machine, as well as human errors were excluded. The training was conducted in the morning. After an accommodation period of 90 s, the first odor was presented for 60 s. In parallel, 12 pulses of 100 V for 1.5 s were delivered with an interval of 3.5 s. After 30 s of flushing with fresh air, the second odor was presented for 60 s. For the subsequent group of flies, the order of the two odors was reversed. For measuring 0 h performance flies were tested about 3 minutes after the end of the conditioning. To determine 2h memory performance, flies were transferred to food vials after conditioning and kept at 25°C until the test. For each genotype and condition the biological replicate is N=12.

For long-term aversive memory, groups of flies were trained with six cycles, where each cycle was composed of 60s of the first odor presented simultaneously with 12 pulses of 90V for 1.5 s with an interval of 3.5s followed by 30s of fresh air and 60s of the second odor. The cycles were spaced by 15 minutes of inter-training-intervals. The flies were starved for 24 hours before testing for long-term memory to increase motivation to climb the Y-Maze and make a decision. The Biological replicate is N=7.

	Primer Name	Sequence
Dop1R2		
Cloning Primers	Dop1R2_5' fw_SpeI	GCTGCAG ACTAGT CAGCCACCACA
	Dop1R2_5' rev_SmaI	CCTGA ACCCGGG GATAAACTTTACCATAATGC
	Dop1R2_fragment_fw_AgeI	GTTCG ACCGGT GATTGCATTGTGTTACCAG
	Dop1R2_fragment_rev_BstEII	GGGCTT GGTAACC ACGACGAATCTTGCGTGGACAG
	Dop1R2_3' fw_XhoI	ATT CTCGAGA AAGTATCAACCCACGATGCGTTC
gRNAs	Dop1R2_3' rev_Acc65I	GGCAAC GGTACC AGATGCAGATACCG
	Dop1R2_gRNA_5'	ACGAACTTAAGATAAAGTGTCGG
	Dop1R2_gRNA_3'	GCATCGTGGGCTGGTACTTCCGG
Screening & Sequencing	Dop1R2_Screening_fw	TATCCCTATGACGTCCCGGAC
	Dop1R2_Screening_rev	GACAGGTTGAGTGATGCGCC
	Dop1R2_5'FRT_fw	GGCTACACATCATTTTATGCCAG
	Dop1R2_5'FRT_rev	GTTCTGTGCCTGATTCTGTTC
	Dop1R2_3'FRT_fw	TCCTAACTGGCTTCTCTTCC
	Dop1R2_3'FRT_rev	AGCGCTTAATTCACGAAAGC

Table 2

Primers and gRNAs for generating Dop1R2 conditional knock-out lines.

Appetitive olfactory conditioning

Before appetitive conditioning, groups of 50 to 100 flies with mixed sex were starved for 19 to 21 h in plastic vials containing damp cotton at the bottom. Experiments in which more than half of the flies died, the flies did not move or there were technical problems with the machine, as well as human errors were excluded. The position in the machine and the sequence in which the genotypes were tested were randomized. The training was conducted in the morning. The conditioning protocol consists of a 90-s accommodation period, 120 s of the first odor, 60 s of fresh air followed by 120 s of the second odor. During the first odor, flies are in a conditioning tube lined with filter paper that was soaked in water the day before the experiment and left to dry overnight. For the second odor, flies are transferred to a conditioning tube lined with a filter paper that was soaked with a 1.5 M sucrose (Sigma-Aldrich, Cat# 84100-1KG; CAS Number 57-50-1) solution on the day before and left to dry at RT. After conditioning, flies were either directly tested for STM or put back in starvation vials until the memory test 2 h later. For 24-h memory, flies were fed for 3 h after training before starving them again. One experiment consisted of 2 reciprocal conditionings, in which the odor paired with sucrose was reversed. For each genotype and condition the biological replicate is N=12.

Memory tests

Flies were loaded into a sliding compartment and transferred to a two-arm choice point. Animals were allowed to choose between 3-octanol and 4-methyl-cyclohexanol. After 60 s, flies trapped in both arms were collected separately and counted. Based on these numbers, a preference index was calculated as follows:

- $PREF = ((N_{arm1} - N_{arm2}) / N_{total}) \cdot 100$ the two preference indices were calculated from the two reciprocal experiments. The average of these two PREFs gives a learning index (LI). $LI = (PREF_1 + PREF_2) / 2$.

In case of all Long-term Aversive memory experiments, the Y-Maze protocol was adapted to test flies 24 hours post training. Testing using the Y-Maze was done following the protocol as described in (Mohandasan et al., 2022 [DOI](#)) where flies were loaded at the bottom of 20-minutes odorized 3D-printed Y-Mazes from where they would climb up to a choice point and choose between the two odors. The learning index was then calculated after counting the flies in each odorized vial as follows: $LI = ((N_{CS-} - N_{CS+}) / N_{total}) \cdot 100$. Where N_{CS-} and N_{CS+} are the number of flies that were found trapped in the untrained and trained odor tube respectively.

Sensory Accuracy tests

Flies were tested for their ability to sense the two used odors 3-octanol and 4-methyl-cyclohexanol as well as electric shock. Therefore, the flies were loaded into a sliding compartment and brought to a two-arm choice point. The flies were allowed to freely choose between an arm containing the stimulus and a neutral arm. All experiments were carried out in the dark. Afterward, the flies in each arm were counted and a preference index was calculated.

For testing the odor response, the flies could choose between one of the odors in the same concentration as used for the behavior experiment and the same amount of paraffin oil for 120s.

- Preference index $PI = ((N_{air} - N_{odor}) / N_{total}) \cdot 100$.

For shock response, the flies could freely choose for 60s between a cooper-grid lined tube getting pulses of 100V or a cooper-grid lined tube getting no electric shock.

- Preference index $PI = ((N_{No\ shock} - N_{shock}) / N_{total}) \cdot 100$.

For testing sugar sensitivity, a group of flies was starved for 1 to 21 h in a tube with damp cotton on the bottom. They could choose for 120 s between a tube lined with filter paper that was soaked in 1.5 M sucrose solution the day before or a tube lined with filter paper that was soaked in distilled water the day before.

- Preference index $PI = (N_{\text{sucrose}} - N_{\text{water}})100 / N_{\text{total}}$

Statistical analysis

To compare performance indices between different groups we used One-way ANOVA (ANalysis Of VAriance) with posthoc Tukey HSD (Honestly Significant Difference) Test Calculator for comparing multiple treatments in R with the package multcomp. In the case of two groups, we performed a *t*-test for comparison.

To verify if the data was normally distributed, we used Shapiro-Wilk's test for normal distribution on the statistics software, Prism GraphPad.

We used One-way ANOVA (Analysis of Variance) with posthoc Tukey HSD (Honestly Significant Difference) test when the data was normally distributed. And the non-parametric counterpart, Kruskal-Wallis test when it wasn't, along with Dunn's multiple comparisons test. We also verified if all the groups were significantly different from zero using a one sample *t*-test for normally distributed data and a Wilcoxon's signed-rank test when the data was not normally distributed.

Single-cell transcriptomic data analysis

Single-cell Transcriptomics data from (Davie et al., 2018b) was downloaded from the Gene Expression Omnibus (GEO). The metadata containing the annotations was also downloaded and used to annotate the clusters. The clustering and further transcriptomic analyses were performed using the Seurat (version 5.2.1) package for R (Hao et al., 2024). The clustering was done at 0.5 Seurat resolution. And the umap was computed using the top ten principal components. The LogNormalize method from Seurat was used to calculate log-normalized gene expression values. The venn diagram in Supplementary Figure 6 was created using BioVenn (Hulsen et al., 2008).

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Additional information

Author Contribution

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Reviewer #1 (Public review):

Summary:

In this manuscript the authors present a novel CRISPR/Cas9-based genetic tool for the dopamine receptor dop1R2. Based on the known function of the receptor in learning and memory, they tested the efficacy of the genetic tool by knocking out the receptor specifically in mushroom body neurons. The data suggest that dop1R2 is necessary for longer lasting memories through its action on α/β and α'/β' neurons but is dispensable for short-term memory and thus in γ neurons. The experiments impressively demonstrate the value of such a genetic tool and illustrate the specific function of the receptor in subpopulations of KCs for longer-term memories.

<https://doi.org/10.7554/eLife.99368.2.sa3>

Reviewer #2 (Public review):

Summary:

This manuscript examines the role of the dopamine receptor, Dop1R2, in memory formation. This receptor has complex roles in supporting different stages of memory, and the neural mechanisms for these functions is poorly understood. The authors are able to localize Dop1R2 function to the vertical lobes of the mushroom body, revealing a role in later (presumably middle-term) aversive and appetitive memory. In general the experimental design is rigorous, and statistics are appropriately applied. The manuscript provides a thorough assessment of how Dop1R2 functions within the mushroom bodies to regulate protein-synthesis dependent and independent memory, and provides a valuable new tool for the community.

Strengths:

- (1) The FRT lines generated provide a novel tool for temporal and spatially precise manipulation of Dop1R2 function. This tool will be valuable to study the role of Dop1R2 in memory and other behaviors potentially regulated by this gene.
- (2) Given the highly conserved role of Dop1R2 in memory and other processes, these findings have high potential to translate to vertebrate species.

<https://doi.org/10.7554/eLife.99368.2.sa2>

Reviewer #3 (Public review):

Summary:

Kaldun et al. investigated the role of Dopamine Receptor Dop1R2 in different types and stages of olfactory associative memory in *Drosophila melanogaster*. Dop1R2 is a type 1 Dopamine receptor that can act both through Gs-cAMP and Gq-ERCa²⁺ pathways. The authors first developed a sophisticated tool where tissue-specific knock-out mutants can be generated using Crispr/Cas9 technology in combination with the Gal4/UAS gene-expression toolkit. They direct the K.O. mutation to intrinsic neurons of the main associative memory centre fly brain: the mushroom body (MB). There are three main types of MB-neurons, or Kenyon cells, according to their axonal projections: a/b; a'/b' and g neurons.

Kaldun et al. found that, while not required for short-term memory, dop1R2 is necessary in a/b and a'/b' but not in gamma neurons to display normal appetitive and aversive middle-term (2h) and long-term (24h) memory. These results showcase a compartmentalized role of Dop1R2 in specific neuronal subtypes of the main memory centre of the fly brain for the expression of middle and long-term memories.

The conclusions of this paper are very well supported by the data, and the authors systematically addressed the requirement of a very interesting type of dopamine receptor in both appetitive and aversive memories. These findings are important for the fields of learning and memory and dopaminergic neuromodulation, among others.

Importantly, the authors of this paper produced a tool to generate tissue-specific knock out mutants of dop1R2. Although reports on the requirement of this gene in different memory phases exist, the genetic tools used here represent the most sophisticated approach to induce a loss of function phenotypes in neurons of interest.

Overall, the authors generated a very useful tool to study dopamine neuromodulation in any given circuit when used in combination with the powerful genetic toolkit available in

Drosophila. The reports on this paper confirmed a previously described role of Dop1R2 in the expression of aversive and appetitive LTM providing spatio-temporal resolution and additionally, they mapped these effects to two types of memory neurons in the fly brain, shedding light into the intricate modulation of dopamine in memory circuits.

<https://doi.org/10.7554/eLife.99368.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

In this manuscript, the authors present a novel CRISPR/Cas9-based genetic tool for the dopamine receptor dop1R2. Based on the known function of the receptor in learning and memory, they tested the efficacy of the genetic tool by knocking out the receptor specifically in mushroom body neurons. The data suggest that dop1R2 is necessary for longer-lasting memories through its action on α/β and α'/β' neurons but is dispensable for short-term memory and thus in γ neurons. The experiments impressively demonstrate the value of such a genetic tool and illustrate the specific function of the receptor in subpopulations of KCs for longer-term memories. The data presented in this manuscript are significant.

Reviewer #2 (Public Review):

Summary:

This manuscript examines the role of the dopamine receptor, Dop1R2, in memory formation. This receptor has complex roles in supporting different stages of memory, and the neural mechanisms for these functions are poorly understood. The authors are able to localize Dop1R2 function to the vertical lobes of the mushroom body, revealing a role in later (presumably middle-term) aversive and appetitive memory. In general, the experimental design is rigorous, and statistics are appropriately applied. While the manuscript provides a useful tool, it would be strengthened further by additional mechanistic studies that build on the rich literature examining the roles of dopamine signaling in memory formation. The claim that Dop1R2 is involved in memory formation is strongly supported by the data presented, and this manuscript adds to a growing literature revealing that dopamine is a critical regulator of olfactory memory. However, the manuscript does not necessarily extend much beyond our understanding of Dop1R2 in memory formation, and future work will be needed to fully characterize this reagent and define the role of Dop1R2 in memory.

Strengths:

(1) The FRT lines generated provide a novel tool for temporal and spatially precise manipulation of Dop1R2 function. This tool will be valuable to study the role of Dop1R2 in memory and other behaviors potentially regulated by this gene.

(2) Given the highly conserved role of Dop1R2 in memory and other processes, these findings have a high potential to translate to vertebrate species.

Weaknesses:

(1) The authors state Dop1R2 associates with two different G-proteins. It would be useful to know which one is mediating the loss of aversive and appetitive memory in Dop1R2 knockout flies.

We thank you for the insightful comment. We agree that it would be very useful to know which G-proteins are transmitting Dop1R2 signaling. To that extent, we examined single-cell transcriptomics data to check the level of co-expression of Dop1R2 with G-proteins that are of interest to us. (Figure 1 S1)

Lines 312-325

“Some RNA binding proteins and Immediate early genes help maintain identities of Mushroom body cells and are regulators of local transcription and translation (de Queiroz et al., 2025; Raun et al., 2025). So, the availability of different G-proteins may change in different lobes and during different phases of memory. The G-protein via which GPCRs signal, may depend on the pool of available G-proteins in the cell/sub-cellular region (Hermans, 2003)., Therefore, Dop1R2 may signal via different G-proteins in different compartments of the Mushroom body and also different compartments of the neuron. We looked at Gao and Gaq as they are known to have roles in learning and forgetting (Ferris et al., 2006; Himmelreich et al., 2017). We found that Dop1R2 co-expresses more frequently with Gao than with Gaq (Figure 1 S1). While there is evidence for Dop1R2 to act via Gaq (Himmelreich et al., 2017). It is difficult to determine whether this interaction is exclusive, or if Dop1R2 can also be coupled to other G-proteins. It will be interesting to determine the breadth of G-proteins that are involved in Dop1R2 signaling.”

(2) It would be interesting to examine 24hr aversive memory, in addition to 24hr appetitive memory.

This is indeed an important point and we agree that it will complete the assessment of temporally distinct memory traces. We therefore performed the Aversive LTM experiments and include them in the results.

Lines 208-228

“24h memory is impaired by loss of Dop1R2

Next, we wanted to see if later memory forms are also affected. One cycle of reward training is sufficient to create LTM (Krashes & Waddell, 2008), while for aversive memory, 5-6 cycles of electroshock-trainings are required to obtain robust long-term memory scores (Tully et al., 1994). So, we looked at both, 24h aversive and appetitive memory. For aversive LTM, the flies were tested on the Y-Maze apparatus as described in (Mohandasan et al., (2022).

Flipping out Dop1R2 in the whole MB causes a reduced 24h memory performance (Figure 4A, E). No phenotype was observed when Ddop1R2 was flipped out in the γ -lobe (Figure 4B, F). However, similar to 2h memory, loss of Ddop1R2 in the α/β -lobes (Figure 4C, G) or the α'/β' -lobes (Figure 4D, H) causes a reduction in memory performance. Thus, Dop1R2 seems to be involved in aversive and appetitive LTM in the α/β -lobes and the α'/β' -lobes.

Previous studies have shown mutation in the Dop1R2 receptor leads to improvement in LTM when a single shock training paradigm is used (Berry et al., 2012). As we found that it disrupts LTM, we wanted to verify if the absence of Dop1R2 outside the MB is what leads to an improvement in memory. To that extent, we tested panneuronal flip-out of Dop1R2 flies for 6hr and 24hr memory upon single shock using the elav-Gal4 driver. We found that it did not improve memory at both time points (Figure 4 S1). Confirming that flipping out Dop1R2 panneuronally does not improve LTM (Figure 4 S1C) and highlighting its irrelevance in memory outside the MB.”

(3) *The manuscript would be strengthened by added functional analysis. What are the DANs that signal through Dop1R. How do these knockouts impact MBONs?*

We thank you for this question. We indeed agree that it is a highly relevant and open question, how distinct DANs signal via distinct Dopamine receptors. Our work here uniquely focusses on Dop1R2 within the MB. We aim to investigate other DopRs and the connection between DANs in the future using similar approaches.

(4) *Also in Figure 2, the lobe-specific knockouts might be moved to supplemental since there is no effect. Instead, consider moving the control sensory tests into the main figure.*

We thank you for this suggestion and understand that in Figure 2 no significant difference is seen. However, we have emphasized in the text that the results from the supplementary figures are just to confirm that the modifications made at the Dop1R2 locus did not alter its normal function.

Lines 156-162

“We wanted to see if flipping out Dop1R2 in the MB affects memory acquisition and STM by using classical olfactory conditioning. In short, a group of flies is presented with an odor coupled to an electric shock (aversive) or sugar (appetitive) followed by a second odor without stimulus. For assessing their memory, flies can freely choose between the odors either directly after training (STM) or at a later timepoint.

To ensure that the introduced genetic changes to the Dop1R2 locus do not interfere with behavior we first checked the sensory responses of that line”

(5) *Can the single-cell atlas data be used to narrow down the cell types in the vertical lobes that express Dop1R2? Is it all or just a subset?*

This is indeed an interesting question, and we thank you for mentioning it. To address this as best as we could, we analyzed the single cell transcriptomic data from (Davie et al., 2018) and presented it in Figure 1 S1.

Reviewer #3 (Public Review):

Summary:

Kaldun et al. investigated the role of Dopamine Receptor Dop1R2 in different types and stages of olfactory associative memory in Drosophila melanogaster. Dop1R2 is a type 1 Dopamine receptor that can act both through Gs-cAMP and Gq-ERCa2+ pathways. The authors first developed a very useful tool, where tissue-specific knock-out mutants can be generated, using Crispr/Cas9 technology in combination with the powerful Gal4/UAS gene-expression toolkit, very common in fruit flies.

They direct the K.O. mutation to intrinsic neurons of the main associative memory centre fly brain-the mushroom body (MB). There are three main types of MB-neurons, or Kenyon cells, according to their axonal projections: a/b; a'/b', and g neurons.

Kaldun et al. found that flies lacking dop1R2 all over the MB displayed impaired appetitive middle-term (2h) and long-term (24h) memory, whereas appetitive short-term memory remained intact. Knocking-out dop1R2 in the three MB neuron subtypes also impaired middle-term, but not short-term, aversive memory.

These memory defects were recapitulated when the loss of the dop1R2 gene was restricted to either a/b or a'/b', but not when the loss of the gene was restricted to g

neurons, showcasing a compartmentalized role of Dop1R2 in specific neuronal subtypes of the main memory centre of the fly brain for the expression of middle and long-term memories.

Strengths:

(1) The conclusions of this paper are very well supported by the data, and the authors systematically addressed the requirement of a very interesting type of dopamine receptor in both appetitive and aversive memories. These findings are important for the fields of learning and memory and dopaminergic neuromodulation among others. The evidence in the literature so far was generated in different labs, each using different tools (mutants, RNAi knockdowns driven in different developmental stages...), different time points (short, middle, and long-term memory), different types of memories (Anesthesia resistant, which is a type of protein synthesis independent consolidated memory; anesthesia sensitive, which is a type of protein synthesis-dependent consolidated memory; aversive memory; appetitive memory...) and different behavioral paradigms. A study like this one allows for direct comparison of the results, and generalized observations.

(2) Additionally, Kaldun and collaborators addressed the requirement of different types of Kenyon cells, that have been classically involved in different memory stages: *g* KCs for memory acquisition and *a/b* or *a'/b'* for later memory phases. This systematical approach has not been performed before.

(3) Importantly, the authors of this paper produced a tool to generate tissue-specific knock-out mutants of *dop1R2*. Although this is not the first time that the requirement of this gene in different memory phases has been studied, the tools used here represent the most sophisticated genetic approach to induce a loss of function phenotypes exclusively in MB neurons.

Weaknesses:

(1) Although the paper does have important strengths, the main weakness of this work is that the advancement in the field could be considered incremental: the main findings of the manuscript had been reported before by several groups, using tissue-specific conditional knockdowns through interference RNAi. The requirement of *Dop1R2* in MB for middle-term and long-term memories has been shown both for appetitive (Musso et al 2015, Sun et al 2020) and aversive associations (Plačaiš et al 2017).

Thank you for this comment. We believe that the main takeaway from the paper is the elegant tool we developed, to study the role of *Dop1R2* in fruit flies by effectively flipping it out spatio-temporally. Additionally, we studied its role in all types of olfactory associative memory to establish it as a robust tool that can be used for further research in place of RNAi knockouts which are shown to be less efficient in insects as mentioned in the texts in line 394-398.

“The genetic tool we generated here to study the role of the *Dop1R2* dopamine receptor in cells of interest, is not only a good substitute for RNAi knockouts, which are known to be less efficient in insects (Joga et al., 2016), but also provides versatile possibilities as it can be used in combination with the powerful genetic tools of *Drosophila*.”

(2) The approach used here to genetically modify memory neurons is not temporally restricted. Considering the role of dopamine in the correct development of the nervous system, one must consider the possible effects that this manipulation can have in the establishment of memory circuits. However, previous studies addressing this question restricted the manipulation of *Dop1R2* expression to adulthood, leading to the same

findings than the ones reported in this paper for both aversive and appetitive memories, which solidifies the findings of this paper.

We thank you for this comment and we agree that it would be important to show a temporally restricted effect of Dop1R2 knockout. To assess this and rule out potential developmental defects we decided to restrict the knockout to the post-eclosion stage and to include these results.

Lines 230-250

“Developmental defects are ruled out in a temporally restricted Dop1R2 conditional knockout.

To exclude developmental defects in the MB caused by flip-out of Dop1R2, we stained fly brains with a FasII antibody. Compared to genetic controls, flies lacking Dop1R2 in the mushroom body had unaltered lobes (Figure 4 S2C).

Regardless, we wanted to control for developmental defects leading to memory loss in flip-out flies. So, we generated a Gal80ts-containing line, enabling the temporal control of Dop1R2 knockout in the entire mushroom body (MB). Given that the half-life of the receptor remains unknown, we assessed both aversive short-term memory (STM) and long-term memory (LTM) to determine whether post-eclosion ablation of Dop1R2 in the MB produced differences compared to our previously tested line, in which Dop1R2 was constitutively knocked out from fertilization. To achieve this, flies were maintained at 18°C until eclosion and subsequently shifted to 30°C for five to seven days. On the fifth day, training was conducted, followed by memory testing. Our results indicate that aversive STM was not significantly impaired in Dop1R2-deficient MBs compared to control flies (Figure 4 S3), consistent with our previous findings (Figure 2). However, aversive LTM was significantly impaired relative to control lines (Figure 4 S3), which also aligned with prior observations. These findings strongly indicate that memory loss caused by Dop1R2 flip-out is not due to developmental defects.”

(3) The authors state that they aim to resolve disparities of findings in the field regarding the specific role of Dop1R2 in memory, offering a potent tool to generate mutants and addressing systematically their effects on different types of memory. Their results support the role of this receptor in the expression of long-term memories, however in the experiments performed here do not address temporal resolution of the genetic manipulations that could bring light into the mechanisms of action of Dop1R2 in memory. Several hypotheses have been proposed, from stabilization of memory, effects on forgetting, or integration of sequences of events (sensory experiences and dopamine release).

We thank you for this comment. We agree that it would be interesting to dissect the memory stages by knocking out the receptor selectively in some of them (encoding, consolidation, retrieval). However, our tool irreversibly flips out Dop1R2 preventing us from investigating the receptor’s role in retrieval. Our results show that the receptor is dispensable for STM formation (Figure 2, Figure 4 Supplement 3), suggesting that it is not involved in encoding new information. On the other hand, it is instead involved in consolidation and/or retrieval of long-term and middle-term memories (Figure 3, Figure 4, Figure 5B).

Overall, the authors generated a very useful tool to study dopamine neuromodulation in any given circuit when used in combination with the powerful genetic toolkit available in Drosophila. The reports in this paper confirmed a previously described role of Dop1R2 in the expression of aversive and appetitive LTM and mapped these effects to two specific types of memory neurons in the fly brain, previously implicated in the expression and consolidation of long-term associative memories.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) On the first view, the results shown here are different from studies published earlier, while in the same line with others (e.g. Sun et al, for appetitive 24h memories). For example, Berry et al showed that the loss of dop1R2 impairs immediate memory, while memory scores are enhanced 3h, 6h, and 24h after training. Further, they showed data that shock avoidance, at least for higher shock intensities, is reduced in mutant (damb) flies. All in all, this favors how important it is to improve the genetic tools for tissue-specific manipulation. Despite the authors nicely discussing their data with respect to the previous studies, I wondered whether it would be suitable to use the new tool and knock out dop1R2 panneuronally to see whether the obtained data match the results published by Berry et al.. Further, as stated in line 105ff: "As these studies used different learning assays - aversive and appetitive respectively as well as different methods, it is unclear if Dop1R2 has different functions for the different reinforcement stimulus" I wondered why the authors tested aversive and appetitive learning for STM and 2h memory, but only appetitive memory for 24h.

Thank you for this comment. To that extent, as mentioned above in response to reviewer #2, we included in the results the aversive LTM experiment (Figure 4). Moreover, we performed experiments along the line of Berry et al. using our tool as shown in Figure 4 S1. Our results support that Dop1R2 is required for LTM, rather than to promote forgetting.

(2) Line 165ff: I can't find any of the supplementary data mentioned here. Please add the corresponding figures.

Thank you for pointing this out. In that line we don't refer to any supplementary data, but to the Figure 1F, showing the absence of the HA-tag in our MB knock-out line. We have clarified this in the text (lines 151-153)

(3) I can't imagine that the scale bar in Figure 1D-F is correct. I would also like to suggest to show a more detailed analysis of the expression pattern. For example, both anterior and posterior views would be appropriate, perhaps including the VNC. This would allow the expression pattern obtained with this novel tool to be better compared with previously published results. Also, in relation to my comment above (1), it may help to understand the functional differences with previous studies, especially as the authors themselves state that the receptor is "mainly" expressed in the mushroom body (line 99). It would be interesting to see where else it is expressed (if so). This would also be interesting for the panneuronal knockdown experiment suggested under (1). If the receptor is indeed expressed outside the mushroom body, this may explain the differences to Berry et al.

Thank you for noting this, there was indeed a mistake in the scale bar which we now fixed. Since with our HA-tag immunostaining we could not detect any noticeable signal outside of the MB, we decided to analyze previously existing single cell transcriptomics data that showed expression of the receptor in 7.99% of cells in the VNC and in 13.8% of cells outside the MB (lines 98-100) confirming its sparse expression in the nervous system. The lack of detection of these cells is likely due to the sparse and low expression of the protein. The HA-tag allows to detect the endogenous level of the locus (it is possible that a Gal4/UAS amplification of the signal might allow to detect these cells).

Regarding the panneuronal knockout, we decided to try to replicate the experiment shown in Berry et al. in Figure 4 S1 and found that Dop1R2 is required for LTM.

(4) Related to learning data shown in Figures 2-4, the authors should show statistical differences between all groups obtained in the ANOVA + PostHoc tests. Currently, only an asterisk is placed above the experimental group, which does not adequately reflect the statistical differences between the groups. In addition, I would like to suggest adding statistical tests to the chance level as it may be interesting to know whether, for example, scores of knockout flies in 3C and 3D are different from the chance level.

Many thanks for this correction, we agree with the fact that the way significance scores were shown was not informative enough. We fixed the point by now showing significance between all the control groups and the experimental ones. We also inserted the chance level results in the figure legends.

(5) Unfortunately, the manuscript has some typing errors, so I would like to ask the authors to check the manuscript again carefully.

Some Examples:

Line 31: the the

Line 56: G-Protein

Line 64: c-AMP

Line 68: Dopamine

Line 70: G-Protein (It alternates between G-protein and G-Protein)

Line 76: References are formatted incorrectly

Line 126: Ha-Tag (It alternates between Ha and HA)

Line 248: missing space before the bracket...is often found

Thank you for noticing these errors, we have now corrected the spelling throughout the manuscript.

(6) In the figures the axes are labelled Preference Index (Pref^I). In the methods, however, the calculation formula is defined as "PREF".

We thank you for drawing attention to this. To avoid confusion, we changed the definition in the methods section so that it could be clear and coherent ("Memory tests" paragraph in the methods section).

"PREF = ((N_{arm1} - N_{arm2}) 100) / N_{total} the two preference indices were calculated from the two reciprocal experiments. The average of these two PREFs gives a learning index (LI). LI = (PREF₁ + PREF₂) / 2.

In case of all Long-term Aversive memory experiments, Y-Maze protocol was adapted to test flies 24 hours post training. Testing using the Y-Maze was done following the protocol as described in (Mohandasan et al., 2022) where flies were loaded at the bottom of 20-minutes odorized 3D-printed Y-Mazes from where they would climb up to a choice point and choose between the two odors. The learning index was then calculated after counting the flies in each odorized vial as follows: LI = ((N_{CS-} - N_{CS+}) 100) / N_{total}. Where NCS- and NCS+ are the number of flies that were found trapped in the untrained and trained odor tube respectively.

Reviewer #2 (Recommendations For The Authors):

(1) In Figures 2 and 3, the legends running two different subfigures is confusing. Would be helpful to find a different way to present.

Thank you for your suggestion. We modified how we present legends, placing them vertically so that it is clearer.

(2) Use additional drivers to verify middle and long-term memory phenotypes.

We agree that it would be interesting to see the role of Dop1R2 in other neurons. To that extent, we looked at long term aversive memory in flies where the receptor was panneuronally flipped out, and did not find evidence that suggested involvement of Dop1R2 in memory processes outside the MB. (Figure 4 S1)

(3) Additional discussion of genetic background for fly lines would be helpful.

Thank you for your advice. We have mentioned the genetic background of flies in the key resources table of the methods sections. Additionally, we also included further explanation on how the lines were created and their genetic background (see “Fly Husbandry” paragraph in the methods section).

“UAS-flp;;Dop1R2 cko flies and Gal4;Dop1R2^{cko} flies were crossed back with ;;Dop^{cko} flies to obtain appropriate genetic controls which were heterozygous for UAS and Gal4 but not Dop1R2^{cko}.”

Reviewer #3 (Recommendations For The Authors):

Line 109 states that to resolve the problem a tool is developed to knock down Dop1R2 in a spatial and temporal specific manner- while I agree that this is within the potential of the tool, there is no temporal control of the flipase action in this study; at least I cannot find references to the use of target/gene switch to control stages of development or different memory phases. However the version available for download is missing supplementary information, so I did not have access to supplementary figures and tables.

Thank you for the comment, as mentioned before it would be great to be able to dissect the memory phases. We show in lines 232 – 250 and Figure 4 S3 that the temporally restricted flip-out to the post-eclosion life stage gave us coherent results with the previous findings, ruling out potential developmental defects.

In relation to my comment on the possible developmental effects of the loss of the gene, Figure 1F could showcase an underdeveloped g lobe when looking at the lobe profiles. I understand this is not within the scope of the figure, but maybe a different z projection can be provided to confirm there are no obvious anatomical alterations due to the loss of the receptor.

We understand the doubt about the correct development of the MB and we thank you for your insightful comment. To that extent we decided to perform a FasII immunostaining that could show us the MB in the different lines (Figure 4 S2) and it appears that there are no notable differences in the lobes development in our knockout line.

It seems that the obvious missing piece of the puzzle would be to address the effects of knocking out Dop1R2 in aversive LTM. The idea of systematically addressing different

types of memory at different time points and in different KCs is the most attractive aspect of this study beyond the technical sophistication, and it feels that the aim of the study is not delivered without that component.

We agree and we thank you for the clarification. As mentioned above in response to Reviewer #2, we decided to test aversive LTM as described in lines –208-228, Figure 4, Figure 4 S1.

Some statements of the discussion seem too vague, and I think could benefit from editing:

Line 284 "however other receptors could use Gq and mediate forgetting"- does this refer to other dopamine receptors? Other neuromodulators? Examples?

Thank you for pointing this out. We Agree and therefore decided to omit this line.

Line 289 "using a spaced training protocol and a Dop1R2 line" - this refers to RNAi lines, but it should be stated clearly.

That is correct, we thank you for bringing attention to this and clarified it in the manuscript.

–Lines 329-330

“Interestingly, using a spaced training protocol and a Dop1R2 RNAi knockout line another study showed impaired LTM (Placais et al., 2017).”

The paragraph starting in line 305 could be re-written to improve clarity and flow. Some statements seem disconnected and require specific citations. For example "In aversive memory formation, loss of Dop1R2 could lead to enhanced or impaired memory, depending on the activated signaling pathways and the internal state of the animal...". This is not accurate. Berry et al 2012 report enhanced LTM performance in dop1R2 mutants whereas Placais et al 2017 report LTM defects in Dop1R2 knock-downs, but these different findings do not seem to rely on different internal states or signaling pathways. Maybe further elaboration can help the reader understand this speculation.

We agree and we thank you for this advice. We decided to add additional details and citations to validate our speculation

Lines 350-353

“In aversive memory formation, loss of Dop1R2 could lead to enhanced or impaired memory, depending on the activated signaling pathways. The signaling pathway that is activated further depends on the available pool of secondary messengers in the cell (Hermans, 2003) which may be regulated by the internal state of the animal.”

"...for reward memory formation, loss of Dop1R2 seems to impair memory", this seems redundant at this point, as it has been discussed in detail, however, citations should be provided in any case (Musso 2015, Sun 2020)

Thank you for noting this. We recognize the redundancy and decided to exclude the line.

Finally, it would be useful to additionally refer to the anatomical terminology when introducing neuron names; for example MBON MVP2 (MBON-g1pedc>a/b), etc.

Thank you for this suggestion. We understand the importance of anatomical terminologies for the neurons. Therefore, we included them when we introduce neurons in the paper.

We thank you for your observations. We recognize their value, so we have made appropriate changes in the discussion to sound less vague and more comprehensive.

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